

Glia-mediated gut-brain cytokine signaling couples sleep to intestinal inflammation

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This **important** work by Malita et al. describes a mechanism by which an intestinal infection causes an increase in daytime sleep through signaling from the gut to the blood-brain barrier. Their findings suggest that cytokines upd3 and upd2 produced by the intestine following infection act on glia of the blood brain barrier to regulate sleep by modulating Allatostatin A signaling. The evidence is **compelling** and elegantly performed using the ample *Drosophila* genetic toolbox, making this work appealing for a broad group of neuroscience researchers interested in sleep and gut-brain interactions.

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Abstract

Sickness-induced sleep is a behavior conserved across species that promotes recovery from illness, yet the underlying mechanisms are poorly understood. Here, we show that interleukin-6-like cytokine signaling from the *Drosophila* gut to brain glial cells regulates sleep. Under healthy conditions, this pathway promotes wakefulness. However, elevated gut cytokine signaling in response to oxidative stress – triggered by immune and inflammatory responses in the intestine – induces sleep. The cytokines Unpaired 2 and -3 are upregulated by oxidative stress in enteroendocrine cells and activate JAK-STAT signaling in glial cells, including those of the blood-brain barrier (BBB). This activity maintains elevated sleep during oxidative-stress-induced intestinal disturbances, suggesting that the JAK-STAT pathway in glia inhibits wake-promoting signaling to facilitate sleep-dependent restoration under these conditions. We find that the enteric peptide Allatostatin A (AstA) enhances wakefulness, and during intestinal oxidative stress, gut-derived Unpaired 2/3 inhibits AstA receptor expression in BBB glia, thereby sustaining an elevated sleep state during gut inflammation or illness. Taken together, our work identifies a gut-to-glial communication pathway that couples sleep with intestinal homeostasis and disease, enhancing sleep during intestinal sickness, and contributes to our understanding of how sleep disturbances arise from gastrointestinal disturbances.

Introduction

Sleep is a conserved behavior essential for physical health and mental well-being. This process maintains physiological balance and promotes recovery from illnesses and other stressors.^{1,2} In healthy states, animals exhibit rhythmic periods of wakefulness and activity, but their sleep patterns significantly increase during illness. This adaptive behavior is related to the fundamental role of sleep in the recovery process, allowing the body to conserve energy and allocate resources towards eliminating pathogens and repairing tissue damage. Sleep patterns are generated by neural circuits within the brain. These circuits engage in complex interactions involving diverse brain regions, neurotransmitters, and signaling pathways to regulate the cycles of sleep and wakefulness.^{3,4} For sleep to be effectively modulated during illness, there must be a dynamic interaction between the physiological states of the body's organs and these central sleep-regulatory circuits. However, the signals that mediate this communication and the mechanisms by which they modulate sleep during health and disease remain poorly defined.

Sickness-induced sleep, a behavior conserved across species including mammals and flies,^{5,6} is influenced by cytokines, which are key mediators of immune and inflammatory responses.^{1,2} Cytokines such as Interleukin 1 (IL-1) and tumor necrosis factor alpha (TNFα) are expressed in the healthy mammalian brain in regions that are implicated in sleep regulation, and their circulating levels change during the normal sleep-wake cycle, peaking during the sleep phase. Furthermore, these factors' effects on sleep appear to be dose-dependent, as low levels of IL-1 can enhance sleep, whereas higher doses can inhibit sleep, indicating a dual functionality. Since immune responses alter the expression of these cytokines, they have been hypothesized to act as "somnogens" that promote sleep during times of infection or illness. However, the connection between sleep and immune function is bidirectional, since sleep deprivation in mammals has been linked to increased inflammatory response via IL-1 and TNFα.² In the fruit fly *Drosophila*, sleep deprivation also seems to influence TNFα in astrocyte-like cells to regulate homeostatic sleep responses that enable sleep rebound after deprivation.⁷ However, the effect of these cytokines on sleep has mostly been linked to their central expression and function within the central nervous system (CNS), while the coupling of cytokines produced by peripheral tissues to sleep-regulatory systems within the brain remains poorly understood.

Disorders affecting the gastrointestinal tract can lead to sleep disturbances,⁸ which are also associated with virtually all mental illnesses.⁹⁻¹¹ Conditions including depression, anxiety, and disturbed sleep are frequently observed in individuals with gut inflammation, and the gut microbiome has also been linked to sleep quality and mental health.^{8,12-14} These associations suggest a strong connection between gut health and sleep. For changes in gut status to bring about behavioral changes, the gut must sense its state of health, damage, or presence of pathogens and release signals that lead to altered cellular function within the brain. This gut-to-brain signaling is mediated in large part by hormonal factors released from specialized endocrine cells of the gut, the enteroendocrine cells (EECs).^{15,16} Like the mammalian intestine, the *Drosophila* gut produces numerous diverse hormones from specialized EECs.¹⁷⁻²² Some of these gut hormones are released in response to nutritional intake, and they diet-dependently modulate sleep patterns and arousability through communication with neuroendocrine centers and brain circuits.²³⁻²⁵

In the fly, enteric infection or damage leads to the production of reactive oxygen species (ROS) and the increased expression of the IL-6-like inflammatory cytokines Unpaired 2 and -3 (Upd2/3) in the absorptive enterocytes, a response required for local gut regeneration.^{26,27} These cytokine factors signal through their receptor, Domeless (Dome), to activate the JAK/STAT signaling pathway in target cells, which is important for both immune function and metabolism in flies, demonstrating a conserved function of cytokine action in this species. While the three related

cytokines Upd1, Upd2, and Upd3 all signal through Dome, Upd2 and in particular Upd3 are IL-6-like cytokines mainly triggered by infection and are directly linked with cellular immune responses²⁸⁻³⁰. As in mammals, cytokines are also produced centrally within the *Drosophila* brain, and neuronal Upd1 acts in a leptin-like manner to regulate feeding³¹, a behavior that is also linked with sleep⁴. Peripheral cytokine signaling has also been shown to modulate sleep in this species, where the fat tissue releases Upd2 to reflect adequate nutrition, and this signal modulates sleep³². Furthermore, Unpaired cytokines have been implicated in the modulation of feeding behavior through effects on glial cells³³. Glial cells, including those making up the blood-brain barrier (BBB), have recently gained attention for their role in sleep regulation in both flies and mammals³⁴⁻³⁷. Neurons in the CNS are separated from the circulatory system by the BBB, a selectively semi-permeable cell layer³⁸, which presents a challenge for peripheral hormones to enter and signal to neurons within the brain. However, glial cells within the BBB are ideally positioned to receive and integrate systemic signals from peripheral organs and modulate neuronal function, thereby relaying peripheral information into the brain.

Here, we demonstrate Upd2 and Upd3 cytokine signaling from endocrine EECs in the intestine in *Drosophila*. Our findings show that Upd2/3 signaling from the EECs to BBB glial cells plays a dual role in sleep regulation. Under normal, healthy conditions, EEC-derived Unpaired signaling sustains wakefulness, whereas in response to oxidative stress that leads to gut inflammation, elevated Unpaired signaling instead promotes sleep. Stress-induced EEC-derived Upd2/3 activates the JAK-STAT pathway in glial cells at the blood-brain interface and adjusts sleep through this activation based on intestinal homeostasis and levels of inflammatory signaling from the gut. Our results suggest that gut-derived Unpaired signaling influences sleep regulation through glial gating of wake-promoting AstA-mediated signals, thus linking intestinal health with CNS-dependent behaviors. These results identify a gut-brain connection by which gut disease impacts sleep regulation.

Results

Gut-derived Unpaired cytokine signaling regulates sleep

To investigate whether cytokine signaling from the gut regulates sleep, we silenced the expression of *upd2* and *upd3* in the EECs, which are a principal endocrine cell type in the gut that releases signals with systemic effects and constitutes the functional basis of gut-brain signaling. Using *voilà-GAL4* (a driver that targets all EECs) to drive RNAi in EECs in conjunction with *Tub-GAL80^{ts}* (hereafter referred to as *voilà>*) for temperature-induced RNAi induction exclusively in the adult stage to prevent developmental effects, we observed significant knockdown of the main IL-6 cytokine, *upd3*³⁰, in dissected adult female midguts (**Fig. 1a**), demonstrating that *upd3* is expressed in EECs under normal homeostatic conditions. To eliminate potential neuron-derived phenotypes, we employed *R57C10-GAL80*, a form of *nSyb-GAL80* that effectively inhibits neuronal GAL4 activity, in combination with *Tub-GAL80^{ts}* and *voilà-GAL4^{25,39}*. To evaluate the effectiveness of this temperature-sensitive EEC-specific driver (referred to as *EEC>* hereafter) upon adult-restricted induction, we examined midgut *upd2* and *upd3* transcript levels and observed significant knockdown of both cytokines, which was reproduced with a second independent RNAi line targeting *upd3* (**Fig. 1b**). Importantly, we did not observe any effect of this manipulation on neuronal expression of *upd2* or *upd3*, supporting the specificity of EEC-targeted knockdown (**Fig. S1a**). Additionally, *GFP* expressed under the control of an *upd3-GAL4* driver containing *upd3* enhancer sequences, was apparent in midgut EECs, marked by staining against the EEC fate determinant Prospero (**Fig. 1c**). To further confirm the expression of *upd3* in EECs, we conducted fluorescent *in situ* hybridization targeting *upd3* and *prospero* in adult midguts. This analysis revealed clear co-localization of *upd3* transcripts with *pros*-positive EECs, consistent with

the expression pattern observed using *upd3-GAL4*-driven GFP in these cells (**Fig. S1b**). These results show that EECs of the adult female midgut are a source of Upd2 and Upd3 under normal conditions.

We then explored whether these EEC-derived cytokines govern sleep under such normal conditions, given the known role of cytokines in both healthy and inflammatory states. We found that the knockdown of the two main immune-related cytokines, *upd2* or *upd3*, using *voilà*> increased the amount of time animals spent asleep (defined as a period of inactivity lasting at least five minutes), especially during the day (**Fig. S1c-e**). This phenotype was reproducible using the more restricted *EEC*> driver that includes *R57C10-GAL80*, suggesting that the EEC-specific loss of these cytokines promotes sleep with pronounced effects during the daytime, when females are typically active (**Fig. 1d-f**). This outcome was not attributable to off-target effects (two independent RNAi lines targeting the main IL-6-like cytokine, *upd3*, produced similar phenotypes) or to effects of the RNAi transgenes themselves (**Fig. 1d-f** and S1f-h). Animals lacking EEC-derived Unpaired signaling also exhibited shorter motion bouts (periods of activity; **Fig. S1i**). However, the EEC-specific knockdown of *upd2* or *upd3* did not reduce motion-bout activity (the intensity of activity during wake periods), implying that the lack of gut cytokine signaling did not reduce general activity when the animals were awake (**Fig. S1k**), and these effects do not arise from RNAi transgene insertions themselves (**Fig. S1m**). These observations suggest a direct influence of gut-derived Upd2 and Upd3 on sleep rather than a broader impact on general activity levels. Given that sleep and feeding are mutually exclusive behaviors, we measured feeding. We did not detect significant alterations in feeding behavior as a consequence of *upd2* or *upd3* knockdown in EECs over a 24-hour period using the automated FLIC system (**Fig. S1o**), nor did we observe an effect of EEC-specific *upd3* overexpression on food consumption via a dye assay (**Fig. S1p**). Furthermore, EEC-specific *upd3* knockdown did not affect the animals' metabolic state as reflected in their levels of stored triacylglyceride (TAG) (**Fig. S1q**). Therefore, the sleep phenotype exhibited by animals with EEC-specific *upd2/3* knockdown is not associated with changes in metabolism, appetite, or feeding behavior.

To further explore the role of Upd2 and Upd3 in sleep regulation, we made use of deletion mutations that disrupt *upd3* alone (*upd3 τ 1*) or in combination with *upd2* (*upd2,3 τ 1*). Both mutant lines exhibited a pronounced increase in sleep (**Fig. 1g-i**), with strong effects on daytime sleep, phenocopying the RNAi-mediated knockdown in the EECs. We additionally disrupted the *upd2* or *upd3* genes specifically in the EECs using somatic tissue-specific CRISPR-mediated deletion. EEC-specific CRISPR-mediated knockout of *upd2* or *upd3*, induced by UAS-controlled gRNA pairs designed to excise portions of each gene's coding sequence, led to significantly elevated sleep (**Fig. 1j-l**), further reinforcing these cytokines' role in sleep modulation.

To further dissect the cellular source of gut-derived cytokines regulating sleep, we analyzed the contribution of the two major EEC populations in the adult *Drosophila* midgut, marked by expression of either Allatostatin C (AstC) or Tachykinin (Tk)²⁰. These two molecularly defined groups encompass the vast majority of EECs. We used *AstC-GAL4*²⁵ and *Tk-GAL4*²⁴ drivers, which are knock-in lines carrying *GAL4* inserted at the endogenous *AstC* or *Tk* loci, thereby enabling precise genetic targeting of EECs based on their native hormone expression profile. To restrict *GAL4* activity to the gut and thus avoid effects from neuronal expression, both drivers were combined with *R57C10-GAL80*, generating *AstC^{Gut}-GAL4* (*AstC^{Gut}*>) and *Tk^{Gut}-GAL4* (*Tk^{Gut}*>) drivers. Using these tools, we selectively knocked down *upd2* or *upd3* in either the AstC-positive EECs or the Tk-positive cells. Knockdown of either cytokine in AstC-positive EECs significantly increased sleep (**Fig. 1m-o**), phenocopying the effect observed with knockdown in all EECs (**Fig. 1d-f**). In contrast, knockdown of *upd2* or *upd3* in Tk-positive EECs had no significant effect on sleep (**Fig. 1p-r**). These findings indicate that AstC-positive EECs are a major source of sleep-regulating Unpaired cytokines, whereas Tk-positive EECs do not appear to contribute significantly to this function. Consistent with this, we also observed effective knockdown of *upd3* transcripts in dissected midguts using the *AstC^{Gut}*> driver, indicating that *upd3* is endogenously expressed in the

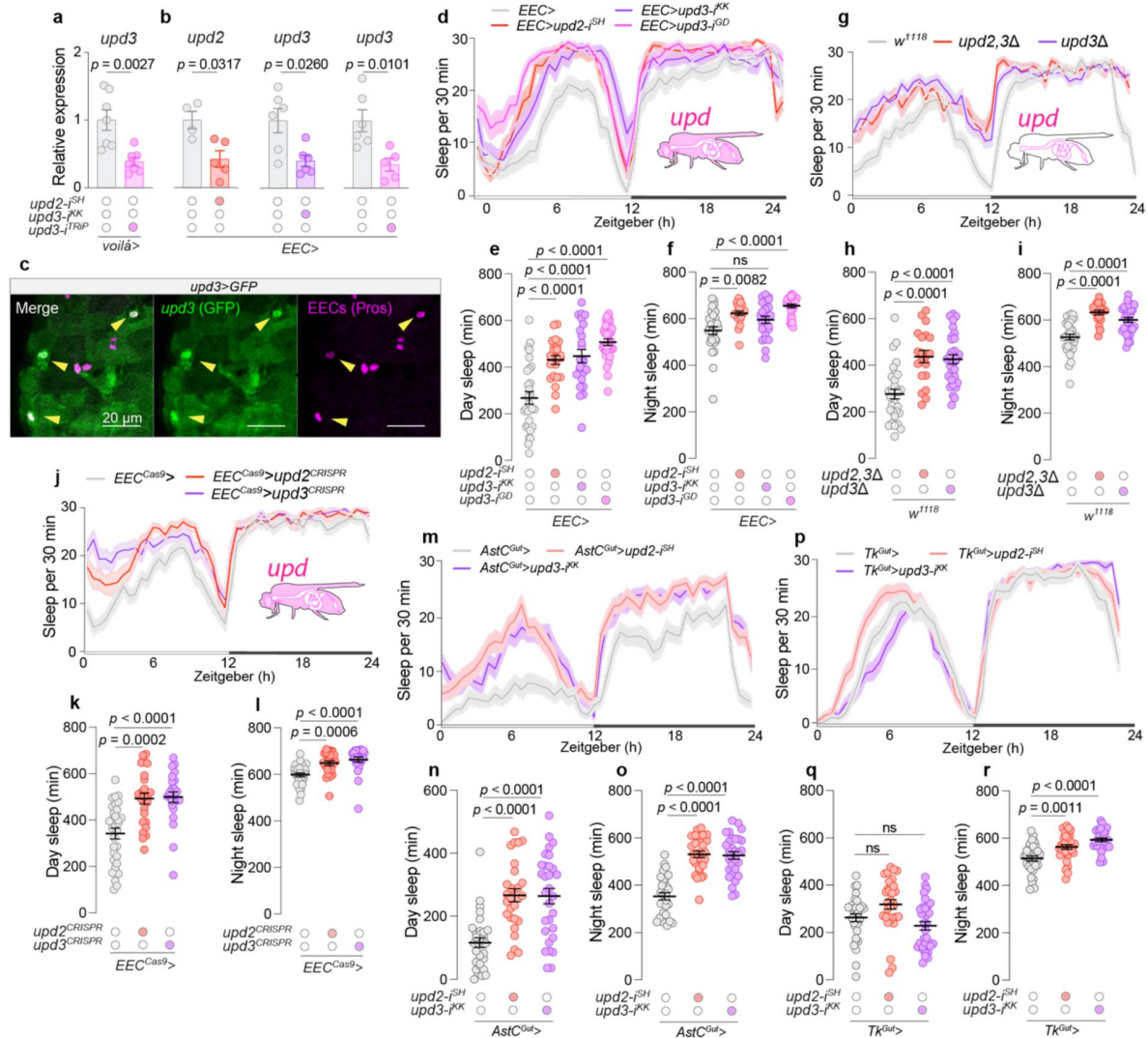


Figure 1.

EEC-derived Unpaired signaling regulates sleep.

a, *upd3* expression levels in midguts expressing RNAi-mediated knockdown of *upd3* in EECs using *voilà*-GAL4 in combination with *Tubulin-GAL80^{ts}* (*voilà*>) (N=7). **b**, *upd2* and *upd3* expression levels in midguts in animals with EEC knockdown of *upd2* or *upd3* using *voilà*-GAL4 in combination with *Tubulin-GAL80^{ts}* and *R57C10-GAL80* (*EEC*>) (N=4-6). **c**, Confocal imaging of *upd3*-GAL4-driven *UAS-GFP* expression in EECs, co-stained with Prospero (Pros) as an EEC marker (Scale bar, 20 μ m). **d**, Twenty-four-hour sleep profiles in animals with EEC-specific *upd2* or *upd3* knockdown (N=29-34). **e**, Total daytime sleep and **f**, nighttime sleep durations in animals with EEC-specific *upd2*- or *upd3*-knockdown flies (N=29-34). **g**, Twenty-four-hour sleep profiles for global *upd2/3* deletion mutants (N=19-32). **h**, Daytime and **i**, nighttime sleep durations in global *upd2/3* mutants (N=19-32). **j**, Sleep profiles following EEC-specific CRISPR-mediated *upd2* or *upd3* knockout (N=24-31). **k**, Daytime and **l**, night sleep durations in animals with EEC-specific CRISPR-mediated *upd2* or *upd3* knockout (N=25-31). **m**, Twenty-four-hour sleep profiles in animals with AsteC-positive-EEC-specific knockdown of *upd2* or *upd3* using *AsteC-GAL4* combined with *R57C10-GAL80* (*AsteC^{Gut}*>) (N=29-30). **n**, Daytime sleep, and **o**, nighttime sleep durations in animals with *AsteC^{Gut}*>-mediated knockdown of *upd2* or *upd3* (N=28-32). **p**, Twenty-four-hour sleep profiles in animals with Tachykinin-positive-EEC-specific knockdown of *upd2* or *upd3* using *Tk-GAL4* combined with *R57C10-GAL80* (*Tk^{Gut}*>) (N=32). **q**, Daytime sleep, and **r**, nighttime sleep durations in animals with *Tk^{Gut}*>-mediated knockdown of *upd2* or *upd3* (N=28-31). Statistical analyses performed using Mann-Whitney tests for panels a and b; ordinary one-way ANOVA with Dunnett's multiple comparisons for panels e, j, h, i, n, o, q, and r; Kruskal-Wallis ANOVA with Dunn's multiple comparisons for panels f, k, and l. Data are presented as mean $\pm$ SEM. ns, non-significant ($p > 0.05$).

AstC-positive EEC population (**Fig. S1r**). Collectively, these findings demonstrate that Upd2 and Upd3 expressed by EECs are important modulators of diurnal sleep patterns in *Drosophila* under normal homeostatic conditions, and they further identify AstC-positive EECs as a key cellular source of these sleep-regulating cytokines.

Glial cytokine JAK-STAT signaling regulates sleep

To identify the CNS targets of EEC-derived Upd2/3 cytokine signaling by which they regulate sleep, we examined the effects of targeted knockdown of the Upd2/3 JAK-STAT-linked receptor *dome* in neurons or glia, the two main cell types in the CNS. Pan-neuronal *dome* knockdown using the driver *R57C10-GAL4*²⁵ did not significantly alter daytime or nighttime sleep in adult females (**Fig. 2a**), thus failing to recapitulate the sleep increase observed upon loss of *upd2* or *upd3* in EECs (**Fig. 1**). This suggests that neurons are not the targets by which gut Unpaired cytokine signaling regulates sleep. In contrast, knockdown of *dome* in glial cells using the pan-glial driver *repo-GAL4* (*repo*>) resulted in a pronounced increase in daytime and nighttime sleep (**Fig. 2a**), similar to the phenotype observed upon EEC-specific loss of Unpaired cytokines. To substantiate this observation further, we silenced *dome* expression in all glia using three independent RNAi lines, all of which strongly induced sleep during the day, effectively ruling out any off-target or transgene-background effects (**Fig. 2b-d** and S2a-e). Additionally, animals with glial *dome* knockdown displayed shorter daytime motion bouts, suggesting reduced periods of wakefulness (**Fig. 2e** and S2f), without exhibiting decreased activity during these bouts (**Fig. 2f** and S2g), phenocopying the effects of *upd2/3* knockdown in EECs (**Fig. S1i,k**). We also assessed whether glial-specific *dome* knockdown might affect feeding and energy storage, but we observed no reduction in food intake (**Fig. S2h**), and no changes in TAG levels were detected (**Fig. S2i**). These findings collectively argue that augmented sleep resulting from impaired JAK-STAT signaling in glia is not due to a general decline in activity but rather represents a specific regulation of sleep itself.

To directly assess the functional Unpaired-mediated communication between the gut and glial cells, we manipulated *upd2* and *upd3* in EECs of animals carrying a ubiquitous transgenic JAK-STAT reporter (*10xSTAT-GFP*)⁴⁰. Knockdown of either *upd2* or *upd3* in the EECs led to a marked decrease in JAK-STAT reporter activity within Repo-positive glial cells under normal conditions (**Fig. 2g,h**), suggesting that Upd2/3-mediated signaling from the EECs to the brain's glial cells activates JAK-STAT signaling. Taken together this suggests that gut-to-glia communication via Upd2 and Upd3 modulates diurnal sleep patterns through glial JAK-STAT activation and that these cytokines are required for the maintenance of wakefulness during the day under healthy conditions.

Oxidative stress modulates sleep through gut-derived cytokine signaling

Having established the significance of gut-derived Unpaired cytokines in maintaining wakefulness under normal conditions, we next explored their role in sleep regulation in the context of gut disturbances that trigger immune and inflammatory responses. Gut infection leads to increased ROS levels and induces local cytokine production^{26,27,41}, and oxidative stress is a key feature of inflammatory conditions in the intestine⁴². Dietary H₂O₂ treatment leads to local intestinal responses comparable to those observed during pathogenic challenges⁴³, suggesting that H₂O₂ feeding provides a controlled method to elevate intestinal ROS levels and examine the specific effects of ROS-induced cytokine signaling. We used this paradigm to ask whether intestinal oxidative stress might elevate the levels of Upd2 and Upd3 in the gut, and we found substantial upregulation of *upd3* expression in dissected midguts of females challenged with ROS by feeding 1% H₂O₂ adult-specific, cornmeal-free diet for 20 hours (**Fig. 3a**). This effect mirrored the upregulation observed with EEC-specific overexpression of *upd3*, indicating that it reflects

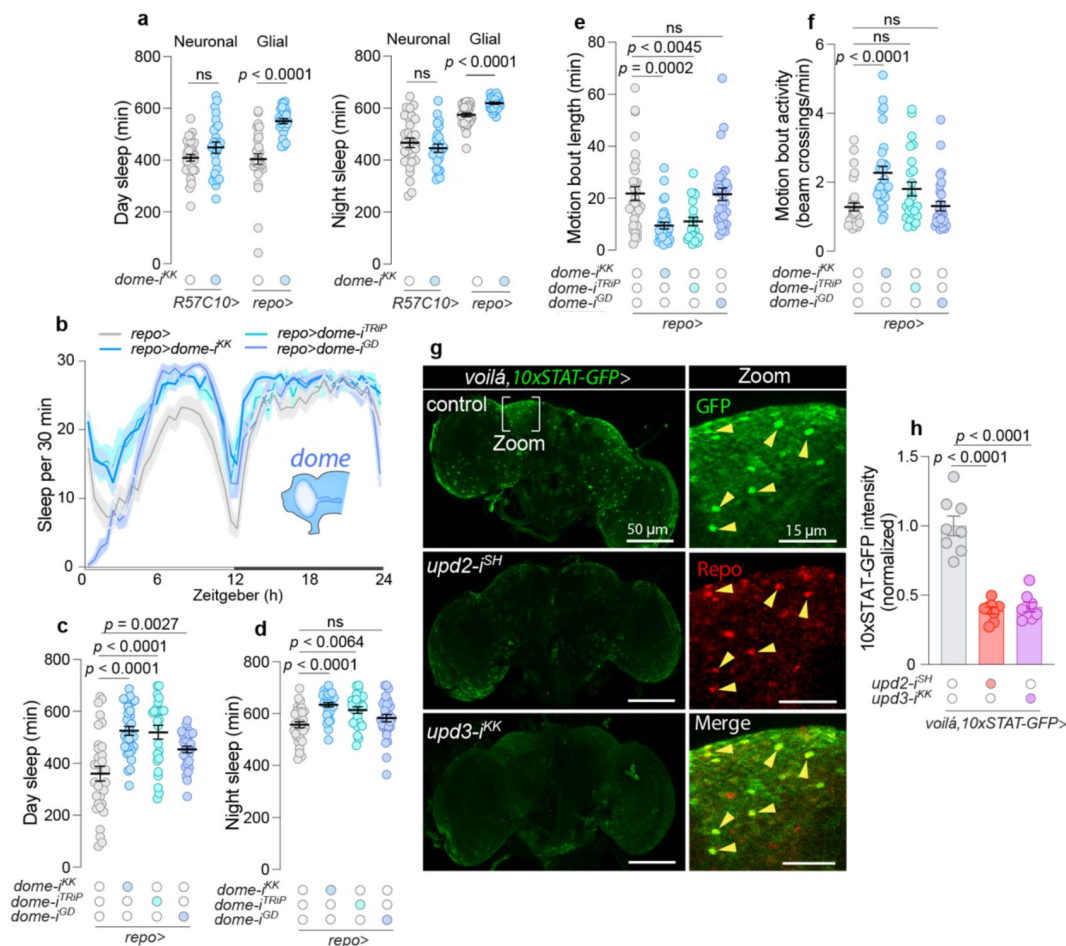


Figure 2.

EEC-derived Unpaired signaling regulates glial JAK-STAT activity that modulates sleep.

a, Day and night sleep measurements for flies with knockdown of IL-6 related Unpaired cytokine receptor *domeless* (*dome*), which activates JAK-STAT, in neurons (*R57C10-GAL4*, *R57C10>*) and glial cells (*repo-GAL4*, *repo>*) (N=25-39). **b**, Twenty-four-hour sleep profiles for controls and animals with glia-specific *dome* knockdown (N=25-32). **c**, Total day sleep duration and **d**, total night sleep duration for animals with glia-specific *dome* knockdown and control flies (N=25-32). **e**, Motion-bout length and **f**, motion-bout activity in animals with glia-specific *dome* knockdown and controls (N=25-32). **g**, Representative images of brains of from controls and animals with EEC knockdown *upd2* or *upd3* using *voilà>*, with *10xSTAT-GFP*. GFP expression (green) reflects JAK/STAT activity, and Repo labeling (red) indicates glial cells (Scale bar, 50 μ m). Insets show zoomed views of STAT-GFP⁺ and Repo⁺ glial cells (Scale bar, 15 μ m). **h**, Quantitative analysis of GFP intensity in the layer of glial cells located at the surface of the brain in animals with EEC knockdown of *upd2* or *upd3* and controls (N=8). Statistical analyses were conducted using parametric *t*-tests for panel a; Kruskal-Wallis ANOVA with Dunn's multiple comparisons for panels c, d, e, and f; and ordinary one-way ANOVA with Dunnett's multiple comparisons for panel h. Data are presented as mean $\pm$ SEM. ns, non-significant ($p>0.05$).

physiologically relevant production of Upd3 by the gut in response to oxidative stress (**Fig. 3a**). Oxidative stress also promoted *upd2* expression, albeit to a lesser extent, and this effect was not modulated by simultaneous EEC-specific *upd3* overexpression.

We next investigated whether sleep is modulated by intestinal oxidative stress and if Unpaired signaling from EEC is required for this response. We induced intestinal oxidative stress by exposing animals to a diet supplemented with H_2O_2 at Zeitgeber Time 0 (ZT0), the onset of the light phase, in a 12-hour light/dark cycle. Exposure to a lower H_2O_2 concentration (0.1%) incrementally increased daytime sleep amount over successive days (**Fig. 3b**). In contrast, a higher H_2O_2 concentration (1%) triggered an immediate augmentation of daytime sleep (**Fig. 3c**). Additionally, to ensure that the observed sleep increase was due to the presence of H_2O_2 itself rather than the procedure of food supplementation, we conducted a control experiment in which animals were fed standard food prepared using the same protocol and replaced daily, but without H_2O_2 . These animals did not exhibit increased sleep, confirming that the effect is attributable to intestinal ROS (**Fig. S3a**).

These observations suggest that intestinal oxidative stress dose-dependently modulates sleep. Since 1% H_2O_2 feeding induced robust responses both in *upd3* expression and in sleep behavior, we asked whether gut-derived Unpaired signaling might be essential for the observed ROS-induced sleep modulation. Indeed, EEC-specific RNAi targeting *upd2* or *upd3* abolished the sleep response to 1% H_2O_2 feeding. Animals with EEC-specific knockdown of *upd2* or *upd3* did not exhibit increased daytime sleep in response to the induction of oxidative stress in the intestine, even over two consecutive days of exposure to 1% H_2O_2 -containing diet (**Fig. 3d,e**). The specificity of this response was corroborated by three independent RNAi lines targeting *upd3*, negating the possibility of RNAi off-target effects (**Fig. 3d,e**), and the loss of response to ROS was also not attributable to the transgenes themselves (**Fig. S3b-e**). Intriguingly, animals lacking *upd3* in the EECs not only did not increase their sleep under oxidative stress but indeed appeared to lose nighttime sleep in response to enteric stress. Moreover, knockdown of *upd2* or *upd3* limited to the AstC-positive EEC subpopulation still prevented the H_2O_2 -induced increase in sleep (**Fig. 3f-h**). These findings indicate that Unpaired signaling from AstC-positive EECs is necessary for mediating the sleep response to intestinal oxidative stress and highlight a specific EEC subtype as a critical source of cytokine signaling in this context.

We next tested whether this sleep phenotype might be associated with general physiological processes rendering animals lacking EEC Unpaired signaling more susceptible to ROS-induced damage. However, when we assessed survival on 1% H_2O_2 -containing food, animals with *upd2* or *upd3* knockdown in EECs displayed no additional sensitivity to oxidative stress, compared to controls (**Fig. 3i**). To further rule out nonspecific toxicity, we examined whether 1% H_2O_2 feeding under our experimental conditions causes neuronal damage. Using a TUNEL assay for apoptosis, we found no evidence of increased neuronal cell death in animals fed 1% H_2O_2 for 24 hours, suggesting that the observed sleep phenotypes are not attributable to general neuronal toxicity (**Fig. S3f,g**). This indicates that the loss of EEC-derived Unpaired signaling specifically leads to an impaired behavioral sleep response to intestinal oxidative stress, rather than to compromised physiological processes that would make the animals more vulnerable to oxidative-stress insults. We also examined whether animals lacking EEC-derived Unpaired signaling exhibit normal behavioral responses to other conditions that modulate sleep, which would suggest a specific requirement for this signaling in responding to intestinal oxidative stress. Animals typically suppress their sleep in response to nutritional deprivation, a behavior conserved across species that is believed to facilitate food-seeking activities and that is also influenced by EEC-mediated hormone signaling^{25,44}. Animals lacking EEC-derived *upd3* suppressed their sleep to a similar extent as controls in response to starvation, indicating a normal sleep response to nutritional stress (**Fig. S3h-j**).

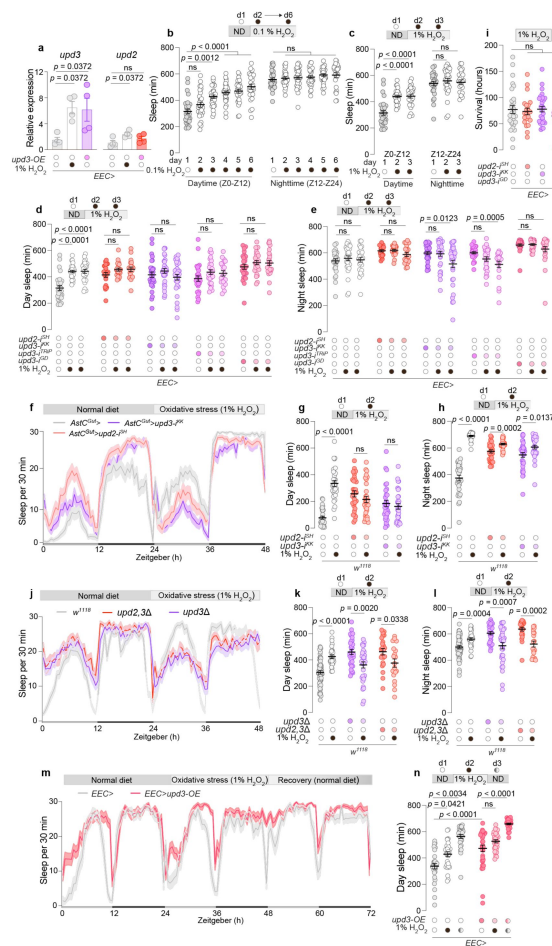


Figure 3.

EEC-derived Unpaired signaling modulates sleep in response to intestinal oxidative stress.

a, Measurement of *upd2* and *upd3* expression in the midgut upon 20-hour treatment with 1% H_2O_2 -laced food or with overexpression of *upd2* (*upd2-OE*) using *EEC>* (N=4). **b** and **c**, Assessment of sleep duration over consecutive days during the daytime (ZT0-ZT12) and nighttime (ZT12-ZT24) in animals exposed to food containing (b) 0.1% H_2O_2 (N=26-31) or (c) 1% H_2O_2 (N=23-29). **d**, Daily sleep, and **e**, nightly sleep amounts measured over one day under standard food conditions followed by two consecutive days on 1% H_2O_2 -containing food in animals with EEC-specific *upd2* or *upd3* knockdown and controls (N=23-30). Experiments measuring sleep levels in controls and animals lacking EEC-derived *upd2* or *upd3* were performed concurrently and share the “control” data, but results are presented in separate figures (b, c, d, and e) for clarity. In **d**, two-way ANOVA revealed significant genotype $\times$ diet interactions for *upd2-i* ($p = 0.0076$), *upd3-i^{KK}* ($p = 0.0003$), *upd3-i^{TRIP}* ($p = 0.0204$), and *upd3-i^{GD}* ($p = 0.0040$), relative to the control, indicating that the sleep response to oxidative stress depends on EEC-derived Unpaired signaling. **f**, Sleep profiles and measurements of daytime (g) and nighttime (h) sleep across a two-day period, encompassing one day on standard diet followed by one day on 1% H_2O_2 -laced food to induce oxidative stress, in flies with *AstC*-positive-EEC-specific knockdown of *upd2* or *upd3* using *AstC^{Gut}>* compared to controls (N=31-32). **i**, Survival rates under a 1% H_2O_2 -induced oxidative stress diet in controls and animals with EEC-specific *upd2* or *upd3* knockdown (N=23-30). **j**, A 48-hour sleep profile comparison between global *upd2/3* mutants and *w¹¹¹⁸* controls under one day of standard food conditions followed by one day of 1% H_2O_2 -induced stress (N=18-63). **k** and **l**, Quantification of daytime and nighttime sleep durations in *upd2/3* mutants versus *w¹¹¹⁸* controls under normal-food conditions and the following day exposed to food containing 1% H_2O_2 (N=18-63). In **k**, two-way ANOVA showed significant genotype $\times$ diet interaction ($p < 0.0001$), confirming a role for Unpaired cytokines in ROS-induced sleep modulation. **m**, Observation of sleep patterns, and **n**, measurements of daytime sleep across a three-day period, encompassing a day on standard diet, subsequent day on 1% H_2O_2 -laced food to induce oxidative stress, and a final day back on standard diet to monitor recovery, in flies with EEC-specific overexpression of *upd3* (*upd3-OE*) compared to controls (N=29-32). Statistical analyses were performed using Kruskal-Wallis ANOVA with Dunn’s multiple comparisons for panels a, b, c, d, e, i, and n; Mann-Whitney test for panels g, h, k, and l. Interaction effects were assessed using two-way ANOVA where indicated. Data are presented as mean $\pm$ SEM. ns, non-significant ($p > 0.05$).

Although we observed behavioral phenotypes with manipulations of either *upd2* or *upd3* alone, suggesting that both are required for normal function, Upd2 and Upd3 likely function at least partially redundantly or additively in their regulation of sleep, as is the case for other processes⁴⁵. Moreover, RNAi effects do not result in a complete loss of function. Therefore, we speculated that a stronger disruption and combined knockout of both *upd2* and *upd3* might lead to even more pronounced phenotypes. We therefore tested the *upd3Δ* single-deletion line and the *upd2,3Δ* double-deletion mutants. Whereas *upd3Δ* and *upd2,3Δ* mutants exhibited increased baseline sleep under homeostatic conditions, these animals not only failed to increase their sleep in response to oxidative stress but indeed showed a strong reduction in daytime and nighttime sleep under oxidative-stress conditions (Fig. 3j–l). These results suggest that contrary to its role in promoting wakefulness during normal homeostatic conditions, the enhanced ROS-induced Unpaired signaling from EECs helps sustain a higher sleep level during periods of oxidative stress. This indicates a dual functionality of Unpaired cytokine signaling, in which low Unpaired signaling promotes wakefulness under normal conditions, whereas higher ROS-induced Unpaired signaling facilitates a shift to restorative sleep during intestinal stress.

We therefore investigated whether higher levels of Unpaired signaling from the gut, comparable to the level produced during oxidative stress, could enhance sleep in the absence of exogenous stressors. We analyzed the effect of *upd3* overexpression in EECs, which drives expression of midgut *upd3* to levels similar to those induced by 1% H₂O₂ feeding (Fig. 3a). Consistent with a model in which high levels of Upd3, like those that would occur during periods of elevated intestinal oxidative stress, promotes daytime sleep, animals overexpressing *upd3* in the EECs exhibited increased sleep during the day, even in the absence of H₂O₂-induced oxidative stress (Fig. 3m,n). These animals further increased their sleep in response to H₂O₂-induced enteric oxidative stress, unlike those lacking gut-derived *upd3* (Fig. 3d,e), suggesting they remain able to mount an additional ROS-induced Unpaired signaling response on top of the overexpression-induced levels. After the animals were switched back to normal food after one day of oxidative stress, both control animals and those with EEC-specific *upd3* overexpression exhibited even more sleep than during the previous day under oxidative-stress conditions (Fig. 3m,n). This suggests a robust recovery sleep response following the insult, likely mediated by Unpaired signaling, since the effect is more pronounced with *upd3* overexpression. Taken together our results show that control animals increase their sleep during oxidative stress, likely as an adaptive recovery response. In contrast, animals with EEC-specific knockdown of *unpaired* cytokines do not exhibit this ROS-induced sleep response; instead, they experience sleep loss under such conditions. This suggests that while Unpaired signaling promotes wakefulness during normal healthy conditions, temporary ROS-induced elevation of gut Unpaired signaling suppresses arousal and leads to more sleep.

EEC-derived Unpaired cytokine signaling activates glial JAK-STAT under oxidative stress

To investigate whether oxidative stress enhances glial JAK-STAT signaling and, if so, whether this enhancement might be mediated by gut-derived Upd2 and Upd3, we assessed glial JAK-STAT reporter activity using the dual-color TransTimer system, which provides temporal information about JAK-STAT signaling⁴⁶. In this system, active STAT promotes the expression of a construct encoding a short-lived destabilized GFP (dGFP, half-life ~2 hours) and a long-lived RFP (half-life ~20 hours) separated by a 2A peptide (6xSTAT-dGFP:2A::RFP); a higher ratio of GFP to RFP in a given cell reflects more recent JAK-STAT signaling. We explored whether JAK-STAT signaling responds dynamically to intestinal oxidative stress and assessed two daily time points. In control animals, we observed no circadian changes between ZT0 (lights on) and ZT12 (lights off) in the superficial layer of cells surrounding the brain (Fig. 4a,b), which is composed of glia⁴⁷. However, we observed a significant increase in GFP signal at ZT0 in animals fed for 20 hours with 1% H₂O₂-containing food, indicating recent JAK-STAT activity in the surface glia. Next, we investigated whether gut-derived Unpaired signaling is responsible for this upregulation by

combining the *10xSTAT-GFP* reporter with knockdown of *upd2* or *upd3* in the EECs. Whereas glial JAK-STAT reporter activity was upregulated by oxidative stress (20 hours' 1%-H₂O₂ feeding) in control animals, this response was abolished in animals with EEC-specific knockdown of *upd2* or *upd3*, indicating that this response is dependent on these EEC-derived cytokines (Fig. 4c,d). Since in this case we used *voilà*> without the pan-neuronal *R57C10-GAL80* element to limit knockdown in EECs, we measured the expression of *upd2* and *upd3* in heads to check for any unintended neuronal effects that might contribute to the observed effect on glial JAK-STAT activity. We detected no changes in the expression of these genes in the head, confirming that the observed JAK-STAT activation in glial cells is attributable to cytokines derived from EECs (Fig. S4a,b). To test the ability of gut-derived Upd3 to drive events in the brain in another way, we made use of cells' homeostatic response to changes in signaling input. Receptor expression is often upregulated in response to low levels of a ligand as a compensatory mechanism to enhance cellular sensitivity⁴⁸. We observed an upregulation of *dome* transcript levels in the heads of animals with EEC-specific knockdown of *upd3*, exposed to oxidative stress induced by 15 hours of feeding with food laced with 1% H₂O₂ (Fig. 4e). Increased *dome* expression suggests reduced Unpaired ligand availability as a result of the loss of EEC-derived Upd3. Together, our results demonstrate that EEC-derived Unpaired cytokine signaling is required for activating glial JAK-STAT under oxidative stress.

Glia JAK-STAT modulates sleep in response to oxidative stress

Since EEC-derived Unpaired signaling promotes oxidative stress-induced sleep and glial JAK/STAT activity, we investigated whether the observed glial JAK-STAT signaling is involved in the modulation of sleep in response to intestinal oxidative stress. Knockdown of *dome* in all glial cells using the *repo*> driver completely abolished the ROS-induced daytime-sleep response when animals were fed 1%-H₂O₂ supplemented diet. The specificity of this effect was confirmed using three RNAi lines and with transgenic RNAi controls (Fig. 5a and S5a-c). Mirroring the effects observed with *upd2,3Δ* mutants (Fig. 3g-i), glia-specific *dome* knockdown ($p < 0.00001$ for *dome-i^{KK}* and $p = 0.0556$ for *dome-i^{TRiP}*) resulted in progressive and substantial sleep loss over two consecutive days on 1% H₂O₂-containing food. To rule out developmental effects, we restricted glial knockdown of *dome* to the adult stage using the *repo*> driver in combination with *Tub-GAL80^{ts}* (*repo^{TS}*>) and observed similar effects (Fig. 5b).

We then examined the dynamics of sleep-regulatory glial JAK-STAT signaling by inducing oxidative stress for one day and then transferring the animals to normal food to observe the recovery response. The results showed that, in response to 1% H₂O₂-containing food, animals lacking glial *dome* expression displayed a sleep response opposite from that of controls, with a reduction in sleep duration rather than an increase, confirmed using independent RNAi lines (Fig. 5c). This phenotype is similar to that seen in the *upd2,3Δ* double mutants (Fig. 3h-j). During the recovery phase, after the animals had been switched back to normal food, the sleep level of controls increased even further, but the sleep duration exhibited by animals with glia-specific *dome* knockdown immediately reverted to pre-stress levels. This pattern indicates that animals with inhibited glial JAK-STAT signaling display an aberrant dynamic sleep response to oxidative stress that is not a consequence of a physiological breakdown but rather arises from altered inhibitory sleep-regulating mechanisms. In line with this and paralleling the loss of *upd2* or *upd3* in the EECs, *dome* knockdown in glial cells did not decrease survival on H₂O₂-containing food (Fig. 5d). This supports the notion that physiological resistance to oxidative stress remains unaltered by gut-glia Unpaired signaling, which in turn indicates that the signaling modulation leads to a specific sleep phenotype. This is likely an important adaptive response under natural conditions, promoting recovery and maintaining homeostasis during or after transient stress episodes.

We further assessed whether glial loss of *dome* affected homeostatic sleep responses induced by sleep deprivation by evaluating the animals' ability to recover sleep after deprivation occurring during the second half of the night (ZT18-ZT24). Like controls, animals with glial-specific *dome*

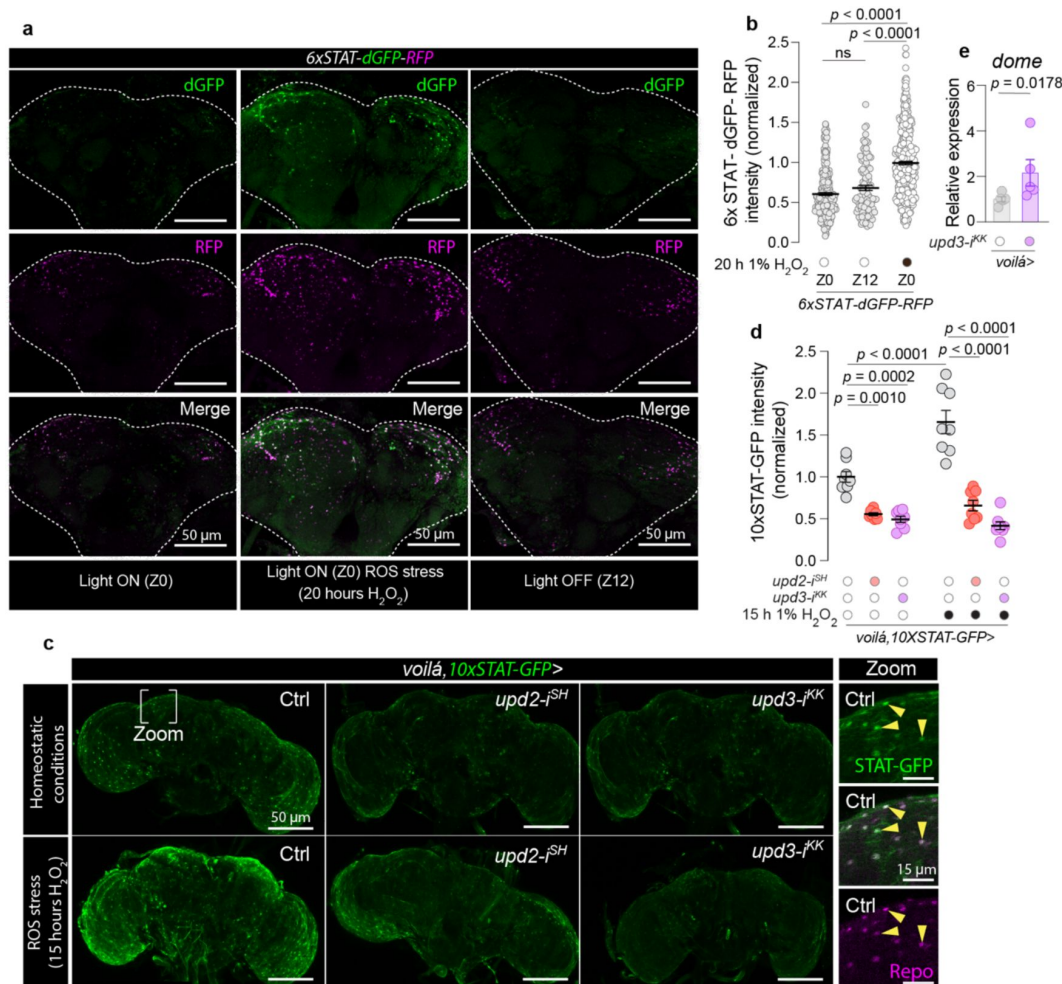


Figure 4.

Activation of glial JAK-STAT signaling by EEC-derived Unpaired cytokines in response to enteric oxidative stress.

a, Representative images of brains from flies expressing the *STAT::dGFP::2A::RFP* reporter, where green (dGFP) reflects recent JAK-STAT activity due to its rapid degradation, and purple (RFP) indicates longer-term pathway activation due to its higher stability. Left panels show brains at lights-on (ZT0), middle panels show brains at lights off (ZT12), and right panels depict brains after 20 hours of oxidative stress induced by 1% H_2O_2 -containing food, imaged at lights-on time ZT0. White dotted lines outline the brain perimeter. Scale bar, 50 μm . **b**, Ratio of dGFP to RFP fluorescence intensity at ZT0, at ZT12, and after oxidative stress (at ZT0), as depicted in panel a, to show dynamic changes in JAK-STAT activity (N=108-461, indicating the number of cells counted). **c**, Representative images of brains displaying *10xSTAT-GFP* expression under homeostatic conditions and after oxidative stress in control flies ("Ctrl") and flies with EEC-specific *upd2* or *upd3* knockdown. Scale bar, 50 μm . Inset panels provide magnified views of glia cells labeled by anti-Repo. Scale bar, 15 μm . **d**, Quantification of *10xSTAT-GFP* intensity in glial cells under homeostatic and oxidative-stress conditions, demonstrating the impact of EEC-specific cytokine knockdown (N=8, indicating the number of brains). **e**, qPCR analysis of *dome* expression in the brains of flies with EEC-specific *upd3* knockdown in comparison to *voilà[>]* controls (N=5). Statistical analyses were conducted using Kruskal-Wallis ANOVA with Dunn's multiple comparisons for panel b; ordinary one-way ANOVA with Tukey's multiple comparisons for panel d; and two-sided unpaired *t*-tests for panel e. Data are presented as mean $\pm$ SEM. ns, non-significant ($p > 0.05$).

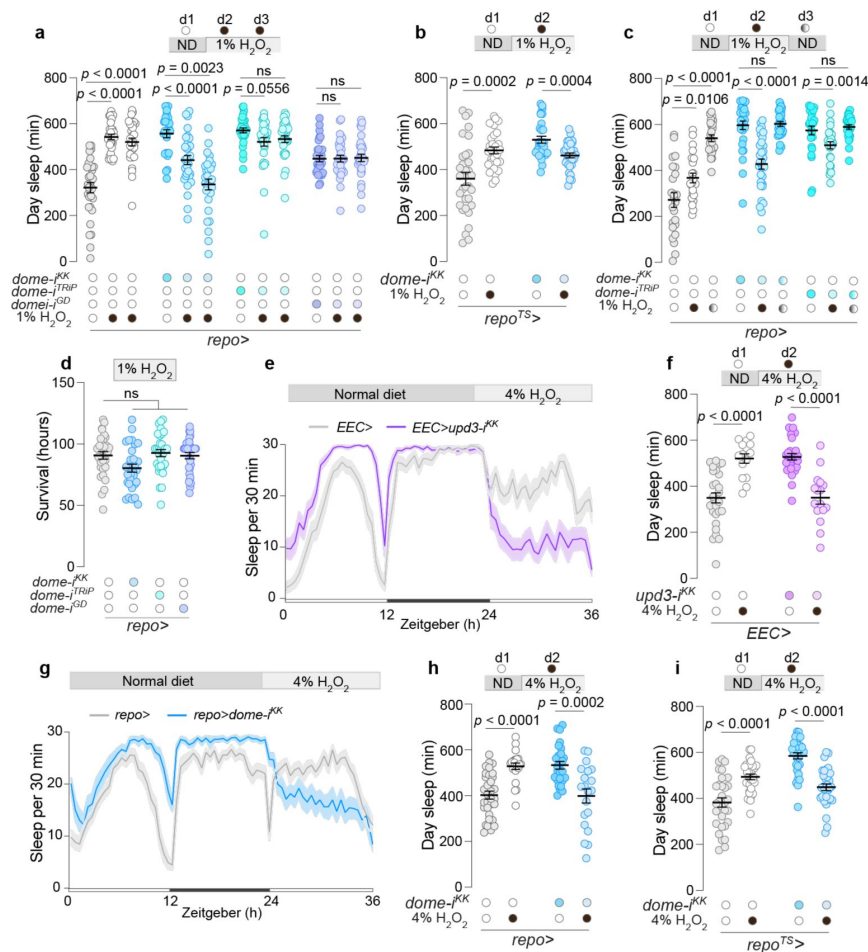


Figure 5.

EEC-derived Unpaired and glial Domeless signaling modulate sleep during intestinal oxidative stress.

a, Daytime sleep duration in flies with glia-specific *dome* knockdown under standard and oxidative-stress conditions induced by 1% H_2O_2 in food (N=25-32). Two-way ANOVA revealed significant genotype $\times$ diet interaction ($p < 0.0001$), indicating that glial Domeless is required for sleep regulation during oxidative stress. **b**, Daytime sleep duration in flies with *repo*-driven *dome* knockdown restricted to the adult stage using *Tub-GAL80^{TS}* (*repo^{TS>}*) under normal conditions and during exposure to 1% H_2O_2 -containing food (N=29-32). Two-way ANOVA showed a significant genotype $\times$ diet interaction ($p < 0.0001$), further supporting a role for glial *dome* in regulating sleep in response to gut oxidative stress. **c**, Daytime sleep during a three-day period, encompassing a day on standard diet, subsequent day on 1% H_2O_2 -laced food to induce oxidative stress, and a final day back on standard diet to monitor recovery, in controls and animals with glia-specific *dome* knockdown (N=24-32). Two-way ANOVA revealed significant genotype $\times$ diet interaction ($p < 0.0001$). **d**, Survival rates of controls and flies with glial-specific *dome* knockdown after exposure to oxidative stress by 1% H_2O_2 -laced food (N=31). **e**, Sleep profiles, and **f**, daytime sleep duration for animals with EEC-specific *upd3* knockdown compared to control flies across a 36-hour period encompassing 24 hours on standard diet followed by 12 hours on oxidative-stress conditions induced by 4% H_2O_2 -containing food (N=15-30). Two-way ANOVA showed a significant genotype $\times$ diet interaction ($p < 0.0001$). **g**, Sleep profiles, and **h**, daytime sleep duration for animals with glia-specific *dome* knockdown compared to control flies across a 36-hour period encompassing 24 hours on standard diet followed by 12 hours under oxidative-stress conditions induced by 4% H_2O_2 -containing food (N=20-32). Two-way ANOVA revealed significant genotype $\times$ diet interaction ($p < 0.0001$). **i**, Nighttime sleep durations for animals under 4% H_2O_2 oxidative-stress conditions in controls and animals expressing adult-restricted knockdown of *dome* in glia (N=31-32). Two-way ANOVA revealed significant genotype $\times$ diet interaction ($p < 0.0001$). Statistical tests: Kruskal-Wallis ANOVA with Dunn's multiple comparisons for panels a, c, and d; Unpaired two-sided *t*-tests for panels b, f, and h; and Mann-Whitney test for panel i. Interaction effects were assessed using two-way ANOVA where indicated. Data are presented as mean $\pm$ SEM. ns, non-significant ($p > 0.05$).

knockdown exhibited increased sleep (rebound sleep) in the morning hours (ZT0-ZT2) following sleep deprivation (**Fig. S5d**,e). This indicates that they exhibit normal rebound sleep responses to deprivation and retain the capability to further increase their sleep. Collectively, these data suggest that Dome-mediated JAK-STAT signaling in the glial cells specifically regulates ROS-induced sleep responses.

We next investigated whether increased intestinal oxidative stress would exacerbate the phenotypes associated with the loss of *upd3* in EECs or *dome* in glial cells by exposing the animals to food containing 4% H₂O₂ and observing changes in their sleep architecture. Oxidative stress resulted in increased sleep in control animals, as anticipated (**Fig. 5e-h**). However, in animals with EEC-specific *upd3* knockdown, or glia-specific *dome* RNAi, exposure to 4% H₂O₂-containing food led to a pronounced loss of sleep during the daytime. For the EEC-specific *upd3* knockdown, the RNAi effect was induced at the adult stage (**Fig. 5e,f**). We therefore also confirmed that adult-restricted knockdown of *dome* in glial cells resulted in similar phenotypes (**Fig. 5i**). Thus, under conditions of intensified intestinal stress induced by 4% H₂O₂ in the food, the loss of *upd3* in EECs phenocopies the glial knockdown of *dome*, leading to reduced sleep and increased wakefulness.

BBB glial JAK-STAT pathway activation drives sleep in response to intestinal oxidative stress

To determine the subset of glial cells responsible for mediating Unpaired-driven sleep regulation, we focused on the perineurial and subperineurial glial cells that form the BBB. These BBB glial cells serve as the interface between the CNS and the periphery, including its organs⁴⁷, and are ideally situated to receive circulating signals from the intestine. Using the *10xSTAT-GFP* reporter, we assessed whether Upd3 from the gut activates JAK-STAT signaling within BBB glial cells. Knockdown of *upd3* in the EECs using *voilà*> (without *R57C10-GAL80*), which in this assay drives specific knockdown in the gut without detectable neuronal effects (**Fig. S4a**,b), resulted in decreased GFP intensity in the outermost glial cell layer of the central brain facing the periphery after 15 hours of exposure to 1% H₂O₂-containing food, indicating reduced JAK-STAT activity in these cells (**Fig. 6a,b**). Interestingly, this knockdown did not affect JAK-STAT activity in the outer glial layer of the ventral nerve cord (VNC), suggesting that Upd3 acts specifically on brain BBB glia in response to intestinal oxidative stress.

We further examined BBB-specific JAK-STAT signaling-mediated effects on sleep by specifically manipulating the subperineurial glial cells – those that form the permeability barrier – using *moody-GAL4* (*moody*>). As observed with other manipulations, knockdown of *dome* in these BBB glial cells led to increased sleep during normal homeostatic conditions (**Fig. 6e,f**). Control animals exhibited the expected sleep increase both during 1%-H₂O₂ exposure and during the recovery period, but loss of *dome* in the subperineurial BBB glia blocked these effects. When oxidative stress levels were elevated further using 4% H₂O₂-containing food, *dome*-RNAi in BBB glial cells led to sleep loss in response to oxidative stress, with sleep levels rebounding to pre-stress levels on the subsequent recovery day when animals were returned to a normal diet (**Fig. 6e,f**). These results indicate that disrupting *dome* specifically in the subperineurial glial cells of the BBB recapitulates the phenotypes observed with pan-glial *dome* knockdown or with EEC-specific *upd3* knockdown, and they suggest that Dome-mediated JAK-STAT activation in the subperineurial BBB cells is *required* for maintaining an increased sleep state during intestinal oxidative stress.

To assess the *sufficiency* of subperineurial JAK-STAT signaling in inducing sleep, we activated the pathway in these cells by expressing a hyperactivated variant of the *Drosophila* JAK ortholog Hopscotch (Hop^{Tum}). Expressing this protein in BBB glia led to increased sleep under normal conditions (without unusual oxidative stress), consistent with a sleep-promoting effect of high JAK-STAT signaling (**Fig. 6c-f**). Moreover, animals with overactive JAK signaling in BBB glia exhibited a further increase in sleep both during oxidative stress and in the subsequent recovery phase, in

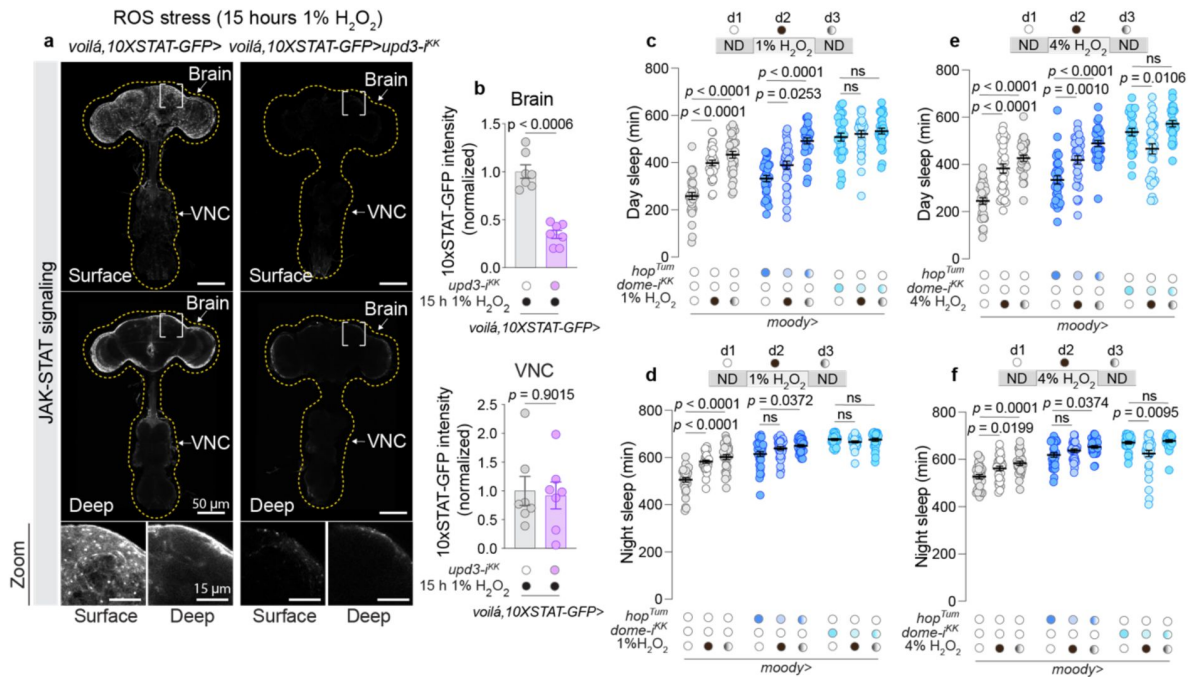


Figure 6.

BBB glia drive Domeless-mediated sleep responses to intestinal oxidative stress.

a, Representative images showing GFP expression driven by *10XSTAT-GFP* in controls and animals with *voilá-GAL4* (*voilá*)-driven *upd3* knockdown in EECs under intestinal ROS induced by 20 hours' exposure to 1%-H₂O₂-containing food. The top panels depict overall brain and ventral nerve cord (VNC) structure with views of surface or deeper layers; the bottom panels provide zoomed-in views, highlighting the BBB glia at the interface between the brain and external environment. Dotted lines indicate brain and VNC perimeters. Scale bars, 50 μ m (top) and 15 μ m (bottom). **b**, Quantification of GFP intensity in the brain and VNC in BBB glia in controls and animals with EEC knockdown of *upd3* under ROS stress, induced by exposure to 1% H₂O₂-laced food (N=7). **c**, Daytime, and **d**, nighttime sleep durations in flies with BBB-glia-specific knockdown of *dome* or overexpression of *hop^{Tum}* under normal conditions, during exposure to 1% H₂O₂-containing food, and subsequent recovery on normal diet (N=23-32). In **c**, two-way ANOVA revealed significant genotype $\times$ diet interaction ($p < 0.0001$), indicating that BBB-glial Domeless is required for daytime sleep regulation under oxidative stress. **e**, Daytime, and **f**, nighttime sleep durations in flies with BBB-glia-specific knockdown of *dome* or overexpression of *hop^{Tum}* under normal conditions, during exposure to 4% H₂O₂-containing food, and subsequent recovery on normal diet (N=23-32). In **e**, two-way ANOVA revealed significant genotype $\times$ diet interaction ($p < 0.0001$), confirming the importance of BBB-glial Domeless signaling during higher levels of oxidative stress. Statistical tests used: Unpaired two-sided *t*-tests for panel b; Kruskal-Wallis ANOVA with Dunn's multiple comparisons in panels c, d and f; ordinary one-way ANOVA with Dunnett's multiple comparisons for panel e. Interaction effects were assessed using two-way ANOVA where indicated. Data are presented as mean $\pm$ SEM. ns, non-significant ($p > 0.05$).

contrast to the effects seen with *dome* knockdown. This suggests that the combined activation of JAK-STAT induced by intestinal ROS and expression of *Hop^{Tum}* leads to additive increases in sleep. Collectively, our data indicate that JAK-STAT signaling specifically in the subperineurial glial of the BBB links sleep responses to intestinal oxidative stress.

AstA signaling promotes wakefulness and mediates ROS-induced sleep regulation in BBB glia

Our results indicate that the effect of gut-to-glia Unpaired cytokine signaling is both dose-dependent and context-dependent. During intestinal oxidative stress, ROS-induced EEC Unpaired signaling leads to high JAK-STAT activity in subperineurial glial cells. Given that animals lacking this gut cytokine-to-glia signaling fail to maintain a high sleep state during oxidative stress, instead exhibiting increased wakefulness, this pathway appears to suppress wake-promoting signals under such conditions. The role of such wake-suppressive effects is likely to enhance sleep, aiding the process of recovery from intestinal damage. To identify potential wake-promoting signals that might be gated by JAK-STAT signaling, we examined a published dataset of genes whose expression in glia is positively or negatively correlated with these cells' JAK-STAT activity following enteric infection³³. The receptors for Allatostatin A (AstA), *AstA-R1* and *AstA-R2*, both ranked among the top 4% of genes most strongly downregulated by JAK/STAT signaling (with *AstA-R1* expression reduced by ~80% and *AstA-R2* by ~90%). Notably, these were the only peptide-hormone G-protein-coupled receptors (GPCRs) downregulated in the JAK-STAT-activate glial cells. This suggests that upon intestinal infection, the JAK-STAT pathway is activated in glial cells, which suppresses AstA signaling by reducing the expression of the AstA receptors. Considering the central role of neuronal AstA in sleep-regulatory circuits^{49,50}, we investigated whether AstA might constitute a wake-promoting signal that is inhibited in glial cells by gut-derived Unpaired signaling. To evaluate the expression pattern of *AstA-R1* and *AstA-R2* within glial populations, we employed *AstA-R1-GAL4* and *AstA-R2-GAL4* knock-in constructs to drive the expression of nuclear-localized *RFP*. We co-stained the brain with antibodies against the glial transcription factor Repo, which marks the nuclei of glial cells. We observed that the outer layer of glial cells at the barrier between the brain and the periphery – constituting the BBB – express both *AstA-R1* and *AstA-R2* (Fig. 7a). These findings are in line with previously reported data showing expression of *AstA-R1* and *AstA-R2* in glial cells referred to above³³. To functionally characterize the role of AstA signaling in these cells, we knocked down *AstA-R2* in BBB glial using *moody*>. This led to a significant reduction in *AstA-R2* transcript levels in dissected brains, indicating that BBB glia is a significant source of *AstA-R2* expression (Fig. S6a). In support of an inhibitory role of Unpaired signaling, we observed that *AstA-R1* and *AstA-R2* expression was upregulated in the heads of animals with EEC-specific *upd3* knockdown, following gut-oxidative stress induced by feeding with 1% H₂O₂-laced food for 20 hours (Fig. 7b). To demonstrate that this is caused by a failure to suppress AstA receptors in glial cells, we examined AstA receptor expression in brains following glia-specific *dome* knockdown in animals fed 1% H₂O₂-containing food for 20 hours. Indeed, glial-specific *dome* knockdown led to strong upregulation of both *AstA-R1* and *AstA-R2*, indicating that ROS-induced glia-mediated Unpaired signaling is inhibiting AstA receptor expression (Fig. 7c).

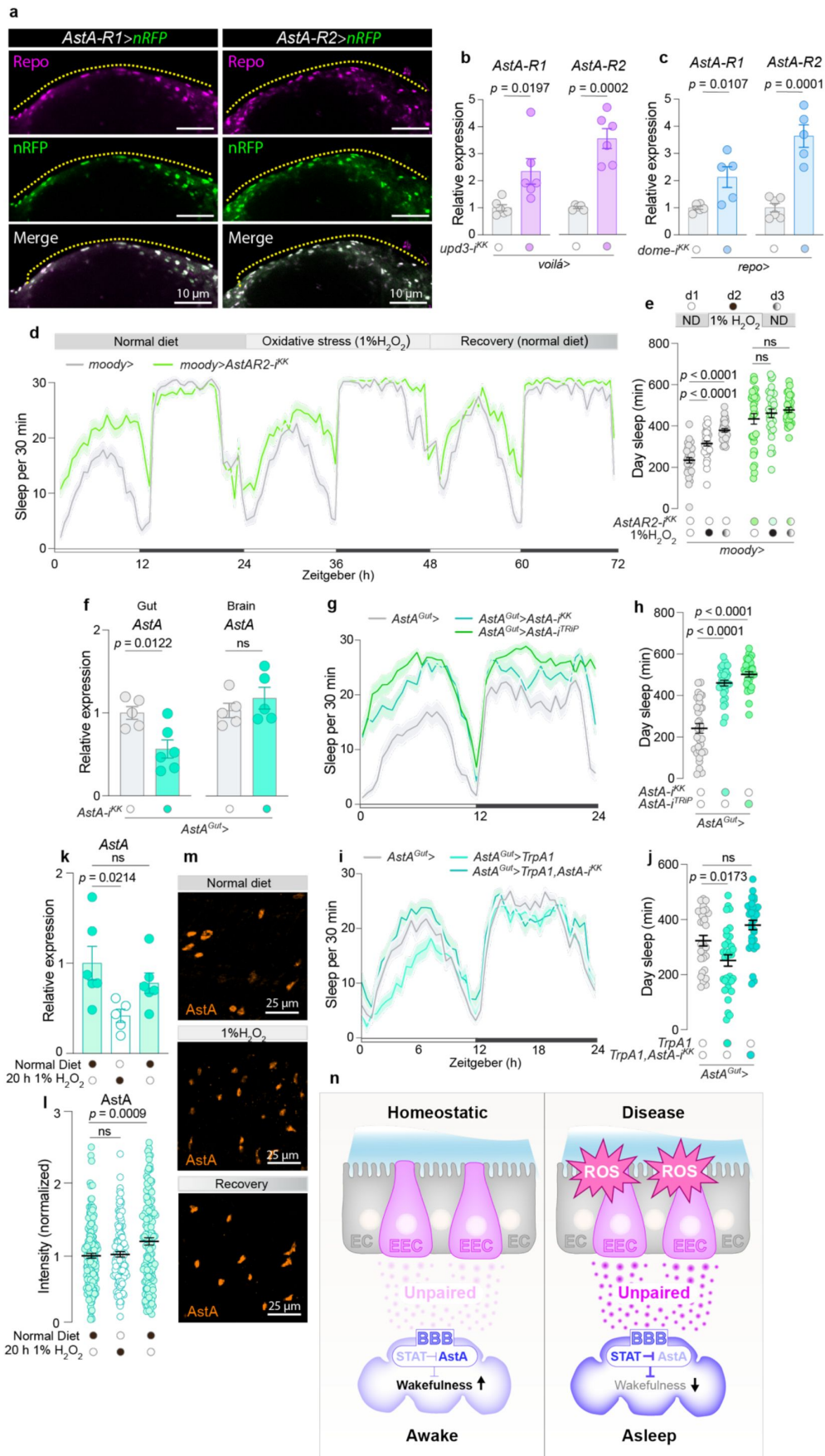


Figure 7.

Gut Unpaired cytokine signaling inhibits wake-promoting AstA signaling.

a, Images of brains from animals with *AstA-R1-GAL4* and *AstA-R2-GAL4* driving nuclear *dsRed* (nRFP, magenta) and co-stained with anti-Repo antibodies (green) show glial cells. The yellow dashed line indicates the interface between the brain and the external space, with the area below housing the (BBB glial cells (Scale bar, 10 μ m). The yellow demarcation accentuates the separation between the cerebral interior and the external milieu, identifying the location of BBB glial cells underneath this partition (Scale bar is 10 μ m). **b**, Relative expression of *AstA-R1* and *AstA-R2* in heads from animals with *upd3* knockdown in EECs driven by *voilà-GAL4* (*voilà>*) compared to the control group after 20 hours on 1% H_2O_2 -laced food to induce oxidative stress (N=5-6). **c**, Relative expression of *AstA-R1* and *AstA-R2* in brains from animals with *dome* knockdown in glial cells driven by *repo-GAL4* (*repo>*) compared to the control group after 20 hours on 1% H_2O_2 -laced food to induce oxidative stress (N=6). **d**, Sleep patterns, and **e**, daytime sleep across a three-day period, encompassing a day on a standard diet, a subsequent day on 1% H_2O_2 -laced food to induce oxidative stress, and a final day back on the standard diet to monitor recovery, in flies with BBB-glia-specific knockdown of *AstA-R2* compared to control (N=28-32). In **e**, two-way ANOVA revealed significant genotype $\times$ diet interaction ($p = 0.0114$, supporting a role for glial *AstA-R2* in ROS-induced sleep regulation). **f**, *AstA* transcript levels in brains and midguts from controls and animals with knockdown of *AstA* in *AstA*⁺ EECs using *AstA-GAL4* (*AstA>*) in combination with *R57C10-GAL80* to suppress neuronal GAL4 activity, referred to as *AstA*^{Gut>} (N=5-6). **g**, Sleep profiles, and **h**, daytime sleep on standard food of animals with *AstA* knockdown in *AstA*⁺ EECs using *AstA*^{Gut>} and controls (N=30-32). **i**, Sleep profiles, and **j**, daytime sleep on standard food of controls, animals with TrpA1-mediated activation of *AstA*⁺ EECs, and animals with TrpA1-mediated activation of *AstA*⁺ EECs with simultaneous knockdown of *AstA* (N=30-32). **k-m**, Quantification of *AstA* transcript levels in whole midguts (**k**) and *AstA* peptide levels in the R5 region of the posterior midgut (**l**) on standard diet, after 1 day on 1% H_2O_2 -laced food to induce oxidative stress, and during recovery following H_2O_2 exposure (**k**: N = 5-6; **l**: N = 126-170). **m**, Representative images of R5 regions stained with anti-AstA antibody (scale bar: 25 μ m). **n**, Diagram illustrating the role of enteroendocrine cells (EECs) in regulating wakefulness and sleep through Unpaired cytokine signaling under homeostatic and stress conditions. Left: Under homeostatic conditions, EECs release baseline levels of Unpaired, which interacts with the blood-brain barrier (BBB) to maintain normal JAK-STAT signaling and *AstA* transduction, promoting wakefulness. Right: In response to stress and disease, reactive oxygen species (ROS) increase in EECs, leading to elevated release of Unpaired. This surge in Unpaired upregulates JAK-STAT signaling in BBB glia, which inhibits wake-promoting *AstA* signaling by suppressing *AstA* receptor expression, thus resulting in increased sleep, a state termed “sickness sleep,” to promote recovery. The diagrams depict the gut lining with EECs highlighted, the interface with the BBB, and the resulting systemic effects on the organism’s sleep-wake states. EC; enterocyte. Statistical tests used: Unpaired two-sided *t*-tests for panel b, c and f; Ordinary ANOVA with Dunnett’s multiple comparisons for panel e, h, j, k, and l. Interaction effects were assessed using two-way ANOVA where indicated. Data are presented as mean $\pm$ SEM. ns, non-significant ($p>0.05$).

Next, we investigated whether *AstA* receptors are involved in mediating the glial-regulated sleep response to intestinal oxidative stress. Like the Unpaired cytokines, *AstA* is released from EECs⁴⁹ and may therefore act as a context-dependent wake-promoting signal that, under certain conditions, is inhibited by Unpaired signaling in BBB glia to promote sleep. We thus hypothesized that during intestinal disturbances characterized by oxidative stress, gut-derived Unpaired signaling via JAK-STAT activation either sustains or consolidates sleep through a mechanism that involves the downregulation of wake-promoting *AstA* receptor signaling in BBB glial cells. In this model, EEC-derived Unpaired signaling normally suppresses *AstA* signaling in BBB glial cells under oxidative stress. Consequently, knocking down *dome* in these cells or *unpaired* in the EECs leads to a failure to downregulate *AstA* receptors, causing the animals to wake up under these conditions. Thus, inhibition of glial *AstA* receptors would impair the animals’ ability to respond to these wake-promoting signals altogether, leaving them unresponsive to intestinal ROS in terms of sleep. Consistent with this notion, we found that knocking down *AstA-R1* or *AstA-R2* in BBB glia attenuated the ROS-induced sleep response (Fig. S6b,c). Knockdown of *AstA-R2* with second,

independent RNAi line resulted in a more pronounced phenotype, with an almost completely blunted sleep response to intestinal ROS, showing no significant sleep increase during oxidative stress or the following day of recovery (**Fig. 7d,e**). AstA-R2 was also the most highly upregulated in response to loss of *upd3* in the EEC or *dome* in glia (**Fig. 7b,c**), and it was the more strongly downregulated of the two AstA receptors in response to glial JAK-STAT activation, together suggesting that AstA-R2 is a primary receptor mediating these effects. Furthermore, knockdown of AstA receptors in BBB glia increased daytime sleep under normal homeostatic conditions, consistent with a wake-promoting role of AstA signaling in BBB glial (**Fig. 7d,e** and S6b,c).

AstA is produced by two cell types, neurons and the EECs in the gut⁴⁹. Since BBB glial cells are well-positioned to receive hormonal signals from the periphery, they likely are regulated by gut-derived AstA. We thus examined whether gut-derived AstA acts as a wake-promoting signal, by conducting *AstA* knockdown in AstA-positive EECs using an *AstA:2A::GAL4* knock-in in combination with *R57C10-GAL80* (*AstA^{Gut}>*) to suppress GAL4 activity in the AstA-positive neuronal population. We confirmed that this driver efficiently reduces the expression of *AstA* in midguts without affecting neuronal *AstA* transcripts levels (**Fig. 7f**). Knockdown of *AstA* in AstA-positive EECs with either of two independent RNAi constructs led to increased sleep without any contribution of the transgenic insertion backgrounds (**Fig. 7g,h** and S6d,e), indicating that gut-derived AstA is indeed a wake-promoting factor. To assess whether EEC-derived AstA is sufficient to promote arousal, we employed the thermosensitive cation channel Transient Receptor Potential A1 (TrpA1)⁵¹ to induce hormonal release from AstA-positive EECs. Activation of these EECs suppressed sleep, an effect that was abolished by simultaneous *AstA* knockdown, supporting that the wake-promoting role of EEC-derived AstA (**Fig. 7i,j**).

Taken together, our findings suggest that enteric oxidative stress induces the release of Unpaired cytokines from the endocrine cells of the gut, which activate the JAK-STAT pathway in subperineurial glia of the BBB surrounding the brain. This activation leads to the glial downregulation of receptors for AstA, which is a wake-promoting factor also released by EECs. Gut-derived Upd signaling thereby gates the effect of AstA at the BBB and permits increased sleep during periods of intestinal stress. We therefore next investigated whether oxidative stress might also regulate the release of AstA from EECs. Following oxidative stress (24 hours of H₂O₂ feeding and the subsequent day), when wild-type animals exhibit increased sleep (**Fig. 7e**), *AstA* transcript levels in the midgut were reduced, accompanied by an accumulation of AstA peptide (**Fig. 7k-m**). This pattern – increased AstC staining in source cells despite decreased expression – suggests that oxidative stress suppresses AstA expression and release. This observation is consistent with a model in which, under conditions of enteric oxidative stress, wake-promoting gut-to-brain AstA signaling is silenced both at the source (gut EECs) and at the target (the BBB glia) by ROS-induced Unpaired signaling. We recently showed that Tk-positive EECs, which make up a population distinct from the AstA-positive EECs, express TrpA1, a ROS-sensitive cation channel known to promote hormone release, and thus exhibit ROS-induced Tk release²⁴. Contrasting with that system, single-cell RNA sequencing data⁵² show that the AstA-expressing EECs do *not* express TrpA1 (**Fig. S6f**). This absence is consistent with our observation that oxidative stress does not promote AstA release (and indeed appears to inhibit it through mechanisms that remain to be explored), reinforcing the idea that gut-derived AstA signaling is actively suppressed rather than stimulated under these conditions. Together, these data support a model in which oxidative stress downregulates wake-promoting AstA signaling in the gut and simultaneously induces Unpaired cytokine signaling, which acts on BBB glia to suppress AstA receptor expression and thus to block the further transduction of wakefulness-promoting AstA signals. This dual-site regulation likely serves to silence arousal signals and promote sleep as a protective response to intestinal stress. This process may aid in recovery and maintain overall organismal homeostasis.

Discussion

Intestinal inflammation and microbial imbalance are strongly associated with sleep disturbances and mental disorders such as anxiety and depression^{8,12,14}. The influence of gut health on CNS-dependent behaviors is thought to be mediated by the gut-brain axis, comprised of diverse signals secreted by the gut that act on the brain to induce behavioral responses⁵³. Whereas the regulation of feeding behavior by this axis has been extensively studied, leading to revolutionary approaches to medical weight loss and diabetes control, the role of gut-brain signaling in regulating sleep – a behavior affected by nearly all illnesses^{54,55} – remains poorly defined. Sickness induces a state of sleepiness, which is believed to be a conserved adaptive response that promotes recovery by supporting energy conservation and efficient immune activity^{5,6}. However, the exact mechanisms driving sickness-induced sleep remain largely elusive. We have demonstrated here that intestinal ROS stress, through driving the release of interleukin 6-like Unpaired cytokines from endocrine cells of the *Drosophila* gut, regulates sleep via a glia-mediated pathway. This gut-to-glia communication promotes sleep during intestinal insult, presumably to facilitate the restorative sleep essential for both physical and mental health. Our findings provide mechanistic insight into how perturbations of gut health can influence sleep, potentially contributing to understanding the link between gastrointestinal disorders, sleep disturbances, and mental illnesses.

Cytokines, key secreted mediators of immune and inflammatory responses, are thought to modulate sleep/wake cycles under disease conditions⁵⁶. Interleukins and TNF α , cytokines induced during illness in mammals, have been suggested to promote sleep to aid recovery from disease. However, most of these effects have been attributed to the actions of cytokines produced within the CNS, leaving open the question of how diseases affecting other parts of the body can drive sleep responses. In *Drosophila*, sleep induced by immune responses is known to be influenced by the NF κ B ortholog Relish in fat tissue⁵⁷, and the neuronally expressed gene *nemuri* drives sleep and connects immune function with sleep regulation⁶. However, inter-organ signaling mechanisms by which intestinal disease or stress regulate sleep have not yet been described in either flies or mammals. Intestinal infection or inflammation leads to elevated levels of ROS in the gut, and our findings demonstrate that enteric oxidative stress in the gut triggers the production of Upd2 and Upd3 cytokines by hormone-secreting EECs. These gut-derived cytokines signal the state of the intestine to brain glial cells, including those of the BBB, and modulate sleep. This glia-mediated gut-to-brain signaling promotes wakefulness in healthy animals under normal conditions, while inducing sleep in response to oxidative stress in the intestine. This indicates a dual functionality, with low levels of gut Unpaired signaling promoting wakefulness and higher stress-induced levels acting to enhance sleep (Fig. 7n). A similar dose-dependent effect has previously been observed for interleukins in rats, in which injection of IL-1 into the CNS can either stimulate sleep or inhibit it, depending on the administered dose⁵⁸. While our findings show that ROS-induced cytokine signaling in the gut modulates sleep through gut-brain communication, an intriguing direction for future research will be to determine whether pathogenic infections – which trigger both intestinal ROS and additional immune pathways – engage distinct, complementary, or overlapping mechanisms compared to chemically induced oxidative stress, and how these immune responses collectively influence sleep regulation.

Our results indicate that Unpaired signaling in subperineurial glial cells – those forming the BBB – activates the JAK-STAT pathway, and they suggest that this effect inhibits wake-promoting AstA signaling by downregulating AstA receptor expression. AstA and its receptors, which are orthologous with the mammalian Galanin signaling system, have been linked to the regulation of sleep, feeding, and metabolism^{49,59,60}. Mammalian glia also express receptors for Galanin⁶¹, which also regulates sleep^{62,63}, further underscoring a conserved role in sleep modulation across species. AstA-producing neurons induce sleep by releasing glutamate onto

sleep-regulatory neuronal circuits, although recent findings also suggest a wake-promoting role for AstA signaling⁵⁰. Irrespective of neuronal AstA, our experiments clearly show that AstA released from EECs of the gut acts as a wake-promoting signal and that activation of AstA receptor signaling in BBB glial cells induces wakefulness. This highlights the potential of peptide hormones to elicit different effects depending on their source tissue and thus their accessible target cells – whether they are produced by the gut outside the BBB or by the CNS inside the barrier. A similar phenomenon has been demonstrated for neuropeptide F^{39,64}. Our findings further suggest that AstA release from EECs is downregulated under oxidative stress in the gut, indicating that this wake-promoting signal is suppressed both at the level of the intestine and at the BBB via Unpaired cytokine signaling. This coordinated downregulation may serve to effectively silence this arousal pathway and promote sleep during intestinal stress.

Interestingly, intestinal ROS can also be generated as a consequence of sleep deprivation^{65,66}, suggesting a potential feedback mechanism. This raises the possibility that ROS produced during sleep loss engages the same Unpaired–JAK-STAT signaling cascade described here, leading to suppression of gut-derived AstA signaling and facilitating recovery sleep. This model provides a mechanistic link between sleep deprivation, intestinal stress, and the regulation of sleep, and suggests that ROS may serve as a physiological signal integrating peripheral stress and behavioral state.

While our study investigated the effects of ROS induction, contrasting findings have been reported under conditions of antioxidant treatment⁶⁵. Our data show both decreased AstA transcript levels and increased AstA peptide accumulation following oxidative stress – a combination typically interpreted as reduced production coupled with peptide retention^{24,25,39}. In contrast, the reported increase in AstA peptide levels under antioxidant treatment was not accompanied by expression data⁶⁵, making it difficult to determine whether the AstA accumulation under these conditions reflects enhanced retention and/or increased production. Furthermore, single-cell RNA sequencing data⁵² indicate that AstA-positive EECs do not express the ROS-sensitive cation channel TrpA1, supporting our observation that intestinal ROS does not stimulate AstA release. We recently found that TrpA1 is expressed in a distinct population of Tk-positive EECs and drives ROS-dependent release of the gut hormone Tk from these cells in *Drosophila*. This mechanism was also observed in the mammalian intestine²⁴. In contrast, a previous report suggested TrpA1-dependent AstA release from EECs⁶⁵, highlighting a potential discrepancy in whether this channel regulates AstA secretion. These differences may reflect context-specific variation in enteroendocrine function, and in any case they underscore the complexity of AstA regulation under varying conditions of gut stress.

Drosophila exhibit conserved behaviors such as sleep, arousal/wakefulness, and anxiety-like responses^{4,67,72}, and the EECs of the fly gut produce diverse hormones similar to those of mammals^{17,22}, potentially influenced by diet, microbiota, and inflammatory responses. This makes *Drosophila* an excellent model for studying behaviors influenced by gut conditions through gut-brain signaling. Our findings suggest that the oxidative-stress level within gut tissues, which is modulated by intestinal bacteria and immune activity^{26,27}, regulates sleep via EEC-derived Unpaired signaling, potentially explaining the observed links between gut microbiota and sleep disturbances in both flies and humans^{13,73}. Furthermore, in mammals, conditions such as inflammatory bowel disease that are linked with oxidative stress⁷⁴ are often associated with sleep and mental-health disturbances^{8,12,14}. Our results imply that cytokines, including interleukin signaling from an inflamed or diseased gut, might be a mechanism by which intestinal illnesses affect sleep and mental health. Our findings raise the possibility that these cytokines may act on glial cells that integrate and relay these gut signals to brain sleep-regulatory circuits.

The neurons of the CNS are isolated from the circulatory system by the BBB⁷⁵ that restricts the transmission of some hormonal and cytokine signals from the periphery to neurons within the brain. Our work suggests that the BBB receives AstA and Unpaired signaling from the periphery.

Other reports indicate that Unpaired cytokines from tumors and from enterocytes also can activate JAK-STAT signaling in BBB glia cells in *Drosophila*^{33,76}. Although our findings highlight endocrine EECs as a primary source of gut-derived cytokines that act on the brain to regulate sleep, it is also possible that enterocytes or other non-endocrine gut cell types contribute to the systemic Unpaired signaling that modulates sleep in response to intestinal oxidative stress. One effect of glial JAK-STAT activity seems to be the alteration of BBB permeability⁷⁶, raising the possibility that EEC-derived Unpaired signaling in BBB glia, directly or through AstA signaling, modulates sleep via regulation of BBB permeability, which has been linked to homeostatic sleep regulation³⁶. Furthermore, the endocytic activity of BBB glia, important for cellular transport and barrier function, has also been associated with sleep regulation³⁷, and thus JAK-STAT-induced changes could regulate sleep through alterations in intracellular trafficking within the cells of the BBB. Another possibility is that JAK-STAT activity might regulate glial metabolic support for neuronal activity and in this way affect sleep patterns. In any case, our findings highlight the involvement of BBB glial cells in transmitting signals from the gut to the brain, adding another layer to our understanding of body-to-brain communication, which suggests that the BBB does more than protect the brain; it also responds to peripheral signals to modulate brain function, presenting an intriguing area for future research into gut-brain signaling.

Methods

Drosophila stocks and husbandry

Flies were cultured using a standard cornmeal-based formulation (82 g/L cornmeal, 60 g/L sucrose, 34 g/L yeast, 8 g/L agar, 4.8 mL/L propionic acid, and 1.6 g/L methyl-4-hydroxybenzoate) maintained at 25 °C with 60% relative humidity under a 12-hour light/dark cycle. Post-eclosion, flies were transitioned to an adult-specific, cornmeal-free diet (comprising 90 g/L sucrose, 80 g/L yeast, 10 g/L agar, 5 mL/L propionic acid, and 15 mL/L of a 10% methyl-4-hydroxybenzoate solution in ethanol)⁷⁷ for 4-7 days prior to experiments. Adult mated females were used for all experiments. Flies were separated by sex one day prior to experimental procedures. Strains harboring the temperature-sensitive *Tubulin-GAL80^{ts}* transgene were initially reared at 18 °C on cornmeal food and then switched to the adult diet for 3-4 days post-eclosion, still at 18 °C. Subsequently, they were incubated at 29 °C for five to seven days to activate RNAi expression in advance of the experiments. To ensure optimal conditions, the flies were provided with fresh food every three days. The following lines used in this study were sourced from the Bloomington *Drosophila* Stock Center (BDSC) at the University of Indiana: *R57C10-GAL4* (#39171); *UAS-upd3-RNAi^{TRiP}* (#32859); *UAS-dome-RNAi^{TRiP}* (#53890); *AstA::2A::GAL4* (#84593); *AstA-R1::2A::GAL4* (#84709); *AstA-R2::2A::GAL4* (#84594); *UAS-AstA-RNAi^{TRiP}* (#25866); *UAS-AstA-R2-RNAi^{TRiP}* (#67864); *UAS-mCD8::GFP* (#5137); *Tub-GAL80^{ts}* (#7108); *repo-GAL4* (#7415); *moody-GAL4* (#90883); *UAS-TrpA1* (#26263); *UAS-dsRed* was extracted from (#8546); *AstC::2A::GAL4* (#84595); *Tk::2A::GAL4* (#84693); and *Upd3-GAL4*, *UAS-GFP* (#98420; GFP variant and protein localization are unknown in this line). Additional fly lines were acquired from the Vienna *Drosophila* Resource Center (VDRC): control line *w¹¹¹⁸* (#60000, which is isogenic with the VDRC RNAi lines); *UAS-upd2-RNAi^{SH}* (#330691); *UAS-upd3-RNAi^{KK}* (#106869); *UAS-upd3-RNAi^{GD}* (#27136); *UAS-dome-RNAi^{KK}* (#106071); *UAS-dome-RNAi^{GD}* (#36356); *UAS-AstA-RNAi^{KK}* (#103215); *UAS-AstA-R1-RNAi^{KK}* (#101395); *UAS-AstA-R2-RNAi^{KK}* (#108648). The *upd3Δ* and *upd2,3Δ* deletion mutants were kindly provided by Bruno Lemaitre. *UAS-upd3* and *UAS-hop^{Tum}* lines were gifts from David Bilder. The *STAT-dGFP;2A::RFP* line was generously supplied by Norbert Perrimon. The *voilà-GAL4* strain was graciously provided by Alessandro Scopelliti. The *R57C10-GAL80* transgene, situated on the X chromosome, was kindly donated by Ryusuke Niwa. The *10xSTAT::GFP* line was a gift from Julien Colombani. To ensure uniformity in genetic background and to create control groups with an appropriate genetic background, all *GAL4* and *GAL80* lines used in this study were backcrossed to a *w¹¹¹⁸* line for multiple generations before being outcrossed with the genetic background specific to the RNAi, CRISPR, or overexpression lines to serve as controls in the experiments²⁵. This

ensures that the only difference between experimental and control animals is the presence or absence of the UAS transgene, providing the most appropriate control for assessing transgene-specific effects.

Generation of tissue-specific CRISPR lines

To facilitate tissue-specific CRISPR-based disruption of the *upd2* and *upd3* loci, constructs were prepared containing two gRNA target sequences, flanked by efficiency-enhancing tRNA sequences. One construct was prepared for *upd2*, and two transgenes, targeting different genomic sites, were made for *upd3*. The *upd2* construct was designed to delete the region encoding the secreted Upd2 protein. One *upd3* construct should delete the initiator ATG codon, and the other – the one used in this work – deletes the second exon, which contains a significant portion of the coding sequence. Target-sequence cassettes were assembled by first cloning the tRNA insert from plasmid pCFD6 (Addgene #73915) between long oligos containing the gRNA target sequences using Q5 polymerase (New England Biolabs, #M0491S). The vector and the PCR products were then integrated using Gibson assembly (NEBuilder HiFi DNA Assembly Master Mix, New England Biolabs, #E2621S). Clones were sequenced to verify accuracy, and correct constructs were integrated into the fly genome at the attP2 site (chromosome 3L) by BestGene (Chino Hills, CA). The sequences used for cloning the *upd2* and *upd3* CRISPR constructs, with gRNA sequences indicated in bold, are shown in **Supplementary Table 1** [↗](#).

Sleep, activity, and survival assays

The *Drosophila* Activity Monitoring System (TriKinetics, Waltham, MA) was employed to track sleep and activity patterns. Single flies aged six to eight days after eclosion were placed into glass tubes using light CO₂ anesthesia. On one end, the tubes were sealed with a foam plug; on the other was placed a detachable 250-μL PCR tube containing 90 μL of feeding medium: either 5% sucrose in 1% agar/water, 5% sucrose mixed with various concentrations of H₂O₂ in 1% agar/water, or plain 1% agar/water for starvation conditions. All food media contained 0.5% propionic acid and 0.15% methyl-4-hydroxybenzoate to prevent microbial growth, with H₂O₂ being supplemented once the food had cooled to below 40 °C. Monitoring of the flies' locomotor activity and sleep began at the beginning of the light cycle, after the animals had spent their first day in the tubes acclimating. Following an additional 24 hours on the standard 5%-sucrose diet, the PCR tubes were replaced with fresh ones containing H₂O₂ or starvation media at the lights-on transition when animals were awake, to avoid unnecessary disturbances to the animals. For recovery experiments, animals were switched back to a 5%-sucrose diet after 24 hours on H₂O₂-laced food. Periods of inactivity lasting 5 minutes or longer were recorded as "sleep." In the sleep deprivation studies, the flies were placed in DAM monitors and subjected to mechanical stimulation, which was produced by attaching the monitors to a vortexer mounting plate (TriKinetics) and vibrating them for 2 seconds at the start of each minute throughout the 6-hour interval leading up to the lights-on time. Recovery sleep was assessed in flies that experienced a reduction of more than 60% in their typical sleep during the deprivation period, using their sleep patterns from the 24-hour period before the onset of sleep deprivation as a baseline. The occurrence of recovery sleep was specifically evaluated during the first two hours immediately following the sleep deprivation phase. For survival assays, flies were loaded into tubes filled with either plain 1% agar/water for starvation or 1% H₂O₂ in 1% agar/water to test oxidative stress resistance. The time of death was recorded upon the complete cessation of movement.

Feeding Assays

Short-term food consumption was quantified using a spectrophotometric dye-feeding assay^{78,79} [↗](#)[↗](#). All food intake experiments were conducted during the time of the normal morning meal, one hour after lights-on in a 12:12-hour dark/light cycle. Flies were transferred without anesthesia to food (90 g/L sucrose, 80 g/L yeast, 10 g/L agar, 5 mL/L propionic acid, and 15 mL/L of a 10% methyl-4-hydroxybenzoate solution in ethanol) supplemented with 0.5%

erionoglucine dye (brilliant blue R, FD&C Blue No.1, Sigma-Aldrich, #861146) and allowed to feed for 1 hour. A control group of flies was provided with undyed food to establish the baseline absorbance levels of fly lysates. For each genotype, 1-2 flies per sample were homogenized in 100 μ L phosphate buffer (pH 7.5) using a TissueLyser LT (Qiagen) with 5-mm stainless-steel beads. Homogenates were centrifuged at 16,000 g for 5 minutes, and 50 μ L of the cleared supernatant was transferred to a 384-well plate. Absorbance was measured at 629 nm for erionoglucine using an EnSight multi-mode plate reader (PerkinElmer). Standard curves for dye were employed to correlate absorbance readings with the amounts of food consumed.

To assess feeding behavior, interactions with food were monitored over a 20-24 hour period using the Fly Liquid-Food Interaction Counter (FLIC) apparatus⁸⁰, *Drosophila* Feeding Monitors (DFMs; Sable Systems) were placed in an incubator set to 25 °C (or 29 °C for strains carrying *GAL80^{ts}*), maintaining 70% humidity under a 12:12-hour light/dark cycle. Each of the 12 DFM chambers was filled with a 10% sucrose solution, and individual flies were introduced in the afternoon following the morning meal. After several hours of acclimation, evening feeding activity was recorded. The following morning, at lights-on, the DFMs were refilled with fresh sugar solution, and data from the morning meal were collected. The feeding behavior was recorded using the manufacturer's software and analyzed using R Studio with the provided package (https://github.com/PletcherLab/FLIC_R_Code).⁸¹

Immunohistochemistry, TUNEL staining, and confocal imaging

Adult midguts, brains, and VNCs were dissected in cold PBS and fixed for 1 hour at room temperature in 4% paraformaldehyde/PBS with gentle shaking. After a quick rinse with PBST (PBS with 0.1% Triton X-100, Merck #12298), the tissues were washed three times for 15 minutes each in PBST. For TUNEL staining, the In Situ Cell Death Detection Kit, Fluorescein (Roche, #11684795910), was used according to the manufacturer's instructions, and tissues were subsequently washed in PBS before mounting. For samples undergoing antibody staining, tissues were then blocked for 30 minutes at room temperature in PBST containing 5% normal goat serum (Sigma) and subsequently incubated overnight (or two days for CNS samples) at 4 °C with primary antibodies diluted in the blocking solution with gentle agitation. After removing the primary antibody solution, tissues were rinsed once and washed three times for 20 minutes each in PBST. Secondary antibodies diluted in PBST were applied, and tissues were incubated overnight at 4 °C, followed by three PBST washes and one PBS wash. The samples were then mounted on poly-L-lysine-coated slides (Sigma, #P8920) in Fluoroshield mounting medium with DAPI (Nordic Biosite, #GTX30920), and imaged on a Zeiss LSM-900 confocal microscope using a 20 $\times$ air or 40 $\times$ oil objective with Zen software. Image stitching was performed using the Stitching function of Zeiss Zen Blue 3.1, and analysis was conducted using the open-source FIJI/ImageJ software package⁸¹. All samples compared with each other within a figure panel were dissected, stained, and imaged simultaneously using identical settings and reagents. For quantification of AstA peptide levels in the R5 region of the posterior midgut, anti-AstA stained images were processed in FIJI, and mean fluorescence intensity was measured in the R5 region. Background signal was subtracted using neighboring non-AstA-expressing areas from the same tissue. To quantify 6xSTAT-dGFP::2A::RFP expression (temporally resolved JAK/STAT activity indicator), dissected brains were fixed for 10 minutes in 4% paraformaldehyde at room temperature with agitation, rinsed once with PBST, mounted, and imaged immediately without antibody staining. For 10xSTAT-GFP quantification, brain and VNC samples were stained as described above, Z-stacks were projected in FIJI using the "sum" method, and Repo-positive cells at the surface were manually segmented to measure the raw integrated density with local background subtraction. For measuring 10xSTAT-GFP across the BBB, linear regions of interest were drawn through the glial layer perpendicular to the brain surface at the plane showing maximum brain size using FIJI's line tool; GFP intensity was then quantified along these lines, and the peak was recorded for each transect. Antibodies used included 1:2000 rabbit anti-AstA (Jena Bioscience, #ABD-062), 1:50 mouse anti-Repo (University of Iowa Developmental Studies Hybridoma Bank, #8D12), 1:20 mouse anti-Prospero (University of Iowa Developmental Studies Hybridoma Bank, #MR1A), 1:500 mouse anti-GFP (ThermoFisher, #A11120), 1:500 chicken

anti-GFP (ThermoFisher, #A10262), 1:1000 rat anti-mCherry (used against Ds Red, ThermoFisher, #M11217), 1:500 Alexa Fluor 488-conjugated goat anti-mouse (ThermoFisher, #A11001), 1:500 Alexa Fluor 555-conjugated goat anti-mouse (ThermoFisher, #A21422), 1:500 Alexa Fluor 647-conjugated goat anti-mouse (ThermoFisher, #A32728), 555-conjugated goat anti-rat (ThermoFisher, #A21434), and 488-conjugated goat anti-chicken (ThermoFisher, #A11039).

In-situ hybridization

To detect co-expression of specific mRNAs, we performed fluorescent *in situ* hybridization using the hybridization chain reaction (HCR) method. This approach, based on a previously published protocol⁸² with slight modifications, utilizes fluorescent probes and reagents from Molecular Instruments (Los Angeles, CA, USA). Adult *Drosophila* midguts were dissected in ice-cold PBS and fixed in 4% paraformaldehyde (PFA) for 1 hour at room temperature with gentle rocking. After fixation, tissues were rinsed and washed three times for 10 minutes each at room temperature in PBS containing 0.1% Tween-20 (Sigma, #P1379). To permeabilize the tissue, samples were incubated at 37 °C for 30 minutes in a buffer containing 1% SDS, 0.5% Tween-20, 50 mM Tris-HCl, 1 mM EDTA, and 150 mM NaCl at pH 7.5. Tissues were then incubated in pre-warmed hybridization buffer (from the HCR buffer kit) for 0 minutes at 37 °C. Hybridization was carried out overnight at the same temperature using hybridization buffer containing 15 nM each of B3-labeled *upd3* probes and B5-labeled *prospero* probes (Molecular Instruments). The next day, tissues were washed four times for 15 minutes each with pre-warmed probe wash buffer, followed by two 5-minute washes in 5× SSCT (prepared using 20× SSC concentrate [Sigma, #S6639] with 0.1% Tween-20 [Sigma, #P1379]). Amplification was initiated by incubating tissues in amplification buffer for 40 minutes at room temperature, followed by an overnight incubation with 120 nM of each corresponding fluorophore-labeled hairpin (Alexa Fluor 488-B3 and Alexa Fluor 546-B5), in the dark at room temperature. After amplification, tissues were washed five times for 10 minutes each in 5× SSCT, followed by six 15-minute washes in PBS. Midguts were mounted on poly-L-lysine-coated slides (Sigma, #P8920) using ProLong Glass Antifade Mountant (Invitrogen, #36984), and coverslipped using a 0.12-mm spacer and 0.1-mm glass coverslip. Imaging was performed with a Zeiss LSM-900 confocal microscope using Zen software.

Triacylglyceride measurements

Triacylglyceride concentrations were determined following established methods^{77,83} using the Randox Triglycerides (GPO-PAP) method (Randox, #TR210). For each sample, flies were homogenized in 50 µl PBS per fly (between 2–4 flies per sample) containing 0.1% Tween-20 (Sigma #1379) using a TissueLyser LT (Qiagen) with 5-mm stainless-steel beads, 50 oscillations/second for 30 seconds. Homogenates were heated at 70 °C for 10 minutes to inactivate endogenous enzymes and centrifuged at 11,000 g for 1 min. Aliquots of cleared and vortexed supernatants (4 µl) were added to 36 µl of triglyceride reagent (Randox, #TR210) in a 384-well plate, covered with ultra-high-clarity optical film (ThermalSeal RT2RR™, Z722553, Excel Scientific). The plate was spun down at 1,500 g for 1 min to settle fluids and eliminate bubbles and incubated for 10 minutes at room temperature. Absorbance for each sample was measured at 540 nm on an Ensign multimode plate reader (PerkinElmer). The readings were then converted to TAG concentrations using standard curves, prepared with triglyceride standards (Randox, 1352TR CAL Standard).

Measurement of transcript levels using qPCR

Several tissue samples containing 3 dissected guts, brains, or heads were collected for each condition or genotype. These samples were then homogenized in 2-ml Eppendorf tubes filled with lysis buffer containing 1% beta-mercaptoethanol, utilizing a TissueLyser LT bead mill (Qiagen) with 5-mm stainless steel beads (Qiagen #69989). RNA extraction was carried out with the NucleoSpin RNA kit (Macherey-Nagel, #740955) following the guidelines provided by the manufacturer. cDNA was synthesized using the High-Capacity cDNA Synthesis kit (Applied Biosystems, #4368814). Quantitative PCR was performed with RealQ Plus 2x Master Mix Green

(Ampliqon, #A324402) using a QuantStudio 5 (Applied Biosystems) instrument. Gene expression results were normalized to the housekeeping gene *Rp49* using the delta-delta-Ct method. The specific oligonucleotides used are given in Supplementary Table 2.

Statistics

Statistical analyses were performed using the Prism software package (GraphPad, version 10). Data were tested for normality before assessments of significance. For data following a normal distribution, pairwise analyses were conducted using two-tailed unpaired Student's *t*-tests, and comparisons involving multiple samples used one-way ANOVA with subsequent post-hoc tests for multiple comparisons. Non-normally distributed data were analyzed using two-tailed unpaired Mann-Whitney *U* tests or one-way Kruskal-Wallis ANOVA, followed by multiple comparisons. Additionally, interactions between genotype and diet were calculated using two-way ANOVA. All plots represent the mean $\pm$ standard error of the mean (SEM). All replicates represent independent biological samples.

Data Availability

All data supporting the findings of this study will be freely available from the lead contact upon request, without restriction.

Code Availability

The MatLAB scripts used for analyzing sleep are described¹⁰[84](#).

Acknowledgements

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Additional information

Author contributions

AM and KR conceived and designed the study. AM, AHS, OK, XC, ECC, NA, TK, and AK performed the experiments and analyzed the data. DSS, KVH, and MJT provided input and contributed to editing the manuscript. AM and KR wrote the manuscript.

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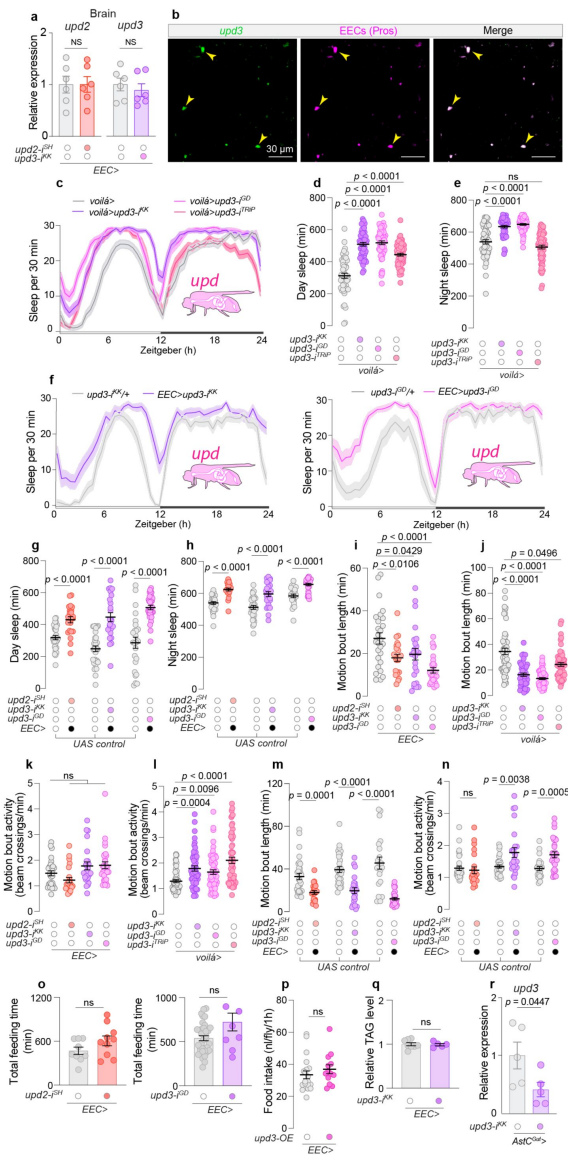


Figure S1.

Effects on sleep, feeding, and metabolic parameters of Unpaired cytokine manipulation in EECs.

a, Expression levels of *upd2* and *upd3* in brains of animals with EEC-specific knockdown using *voilà-GAL4* in combination with *Tubulin-GAL80^{TS}* and *R57C10-GAL80* (together, “EEC>”), showing no significant changes in expression (N = 6). **b**, Representative fluorescent *in situ* hybridization images showing the co-expression of *upd3* and the EEC marker *prospero* in the midgut (Scale bar, 10 μ m). **c**, Twenty-four-hour sleep profiles for animals with EEC knockdown of *upd2* or *upd3* using *voilà-GAL4* (*voilà>*). **d**, Day, and **e**, night sleep durations for EEC *upd2* or *upd3* knockdown and control groups (N=50-61). **f**, Sleep profiles for *upd3* knockdown using *R57C10-GAL80*, *Tub-GAL80^{TS}*, *voilà>* (EEC>) and UAS-*upd3-RNAi* controls (N=23-31). **g**, Day, and **h**, night sleep durations for UAS-RNAi controls and EEC-specific *upd2* or *upd3* knockdown (N=23-31). **i**, **j**, Motion-bout length and **k**, **l**, motion-bout activity for *upd2* or *upd3* knockdown using EEC> or *voilà>* and control flies (N=23-60). **m**, Motion-bout length and **n**, motion-bout activity for EEC *upd2* or *upd3* knockdown and UAS-RNAi control flies (N=21-32). **o**, Total feeding time for EEC-specific *upd2* or -3 knockdown and control flies using FLIC (N=9-36). **p**, Food intake over 1 hour for animals with EEC-specific *upd3* overexpression (*upd3-OE*) and controls (N=13-20). **q**, Relative triacylglyceride (TAG) levels in EEC-specific *upd3* knockdowns and control flies (N=5-8). **r**, Expression levels of *upd3* in midguts of animals with AstC-positive-EEC-specific knockdown of *upd3* using *AstC-GAL4* combined with *R57C10-GAL80* (*AstC^{Gut}>*) (N=5). Statistical analyses were performed using two-sided unpaired *t*-tests for panel a, g, h, q, and r; Mann-Whitney tests for panels m, n, o, and p; ordinary one-way ANOVA with Dunnett’s multiple comparisons panel i; Kruskal-Wallis ANOVA with Dunn’s multiple comparisons for panels d, e, j, k, and l. Data are presented as mean $\pm$ SEM. ns, non-significant ($p > 0.05$).

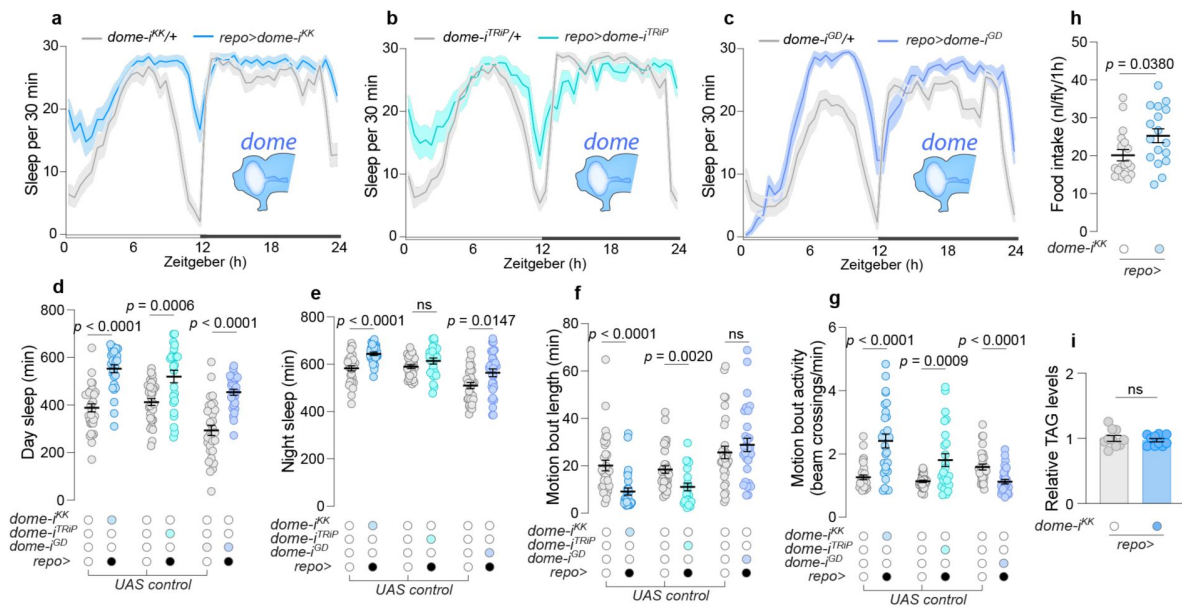


Figure S2.

Characterization of sleep patterns, activity, and metabolic impact of *dome* knockdown in glial cells.

a-c, Sleep profiles over a 24-hour period for animals with glia-specific *dome* knockdown and *UAS-RNAi* controls (N=25-32). **d**, Total daytime sleep, and **e**, nighttime sleep for *UAS-RNAi* controls and animals with glia-specific *dome* knockdown (N=25-32). **f**, Duration of motion bouts and **g**, motion-bout activity for the same groups as in a-d (N=25-32). **h**, Food intake measured over a 1-hour period for animals with *dome* knockdown in glia and *UAS-RNAi* control flies (N=23-31). **i**, Triacylglyceride (TAG) levels relative to controls in glia-specific *dome* knockdown flies. Statistical analyses were performed using Mann-Whitney tests for panels d, e, f, g, h, and i, except t-tests were performed for *dome-1^{TRIP}* versus control in panel d and g and for *dome-1^{GD}* versus control in panel d. Data are presented as mean $\pm$ SEM. ns, non-significant ($p > 0.05$).

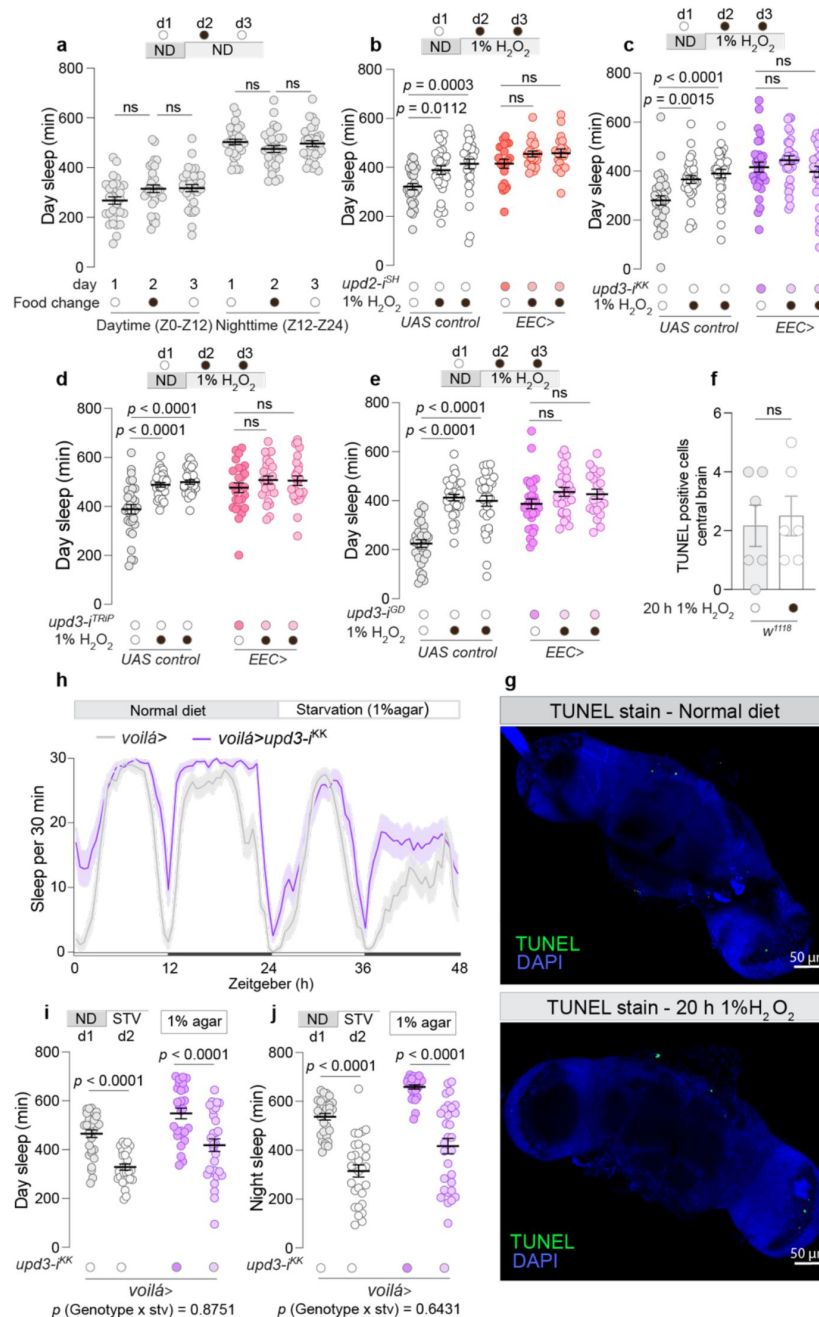


Figure S3.

Sleep duration influenced by EEC-specific Unpaired cytokine disruption and dietary changes.

a, Daytime and nighttime sleep duration in control flies fed standard food prepared and replaced daily without H_2O_2 to control for potential effects of the supplementation procedure (N=32). **b-e**, Daytime sleep duration in flies with *upd2* or *upd3* knockdown in EECs and *UAS-RNAi* controls measured over one day under standard food conditions followed by two consecutive days on 1% H_2O_2 -containing food (N=21-32). **f, g** TUNEL staining of adult *Drosophila* brains following 24-hour feeding with 1% H_2O_2 . TUNEL-positive cells per brain are shown in **f** (N=6), with representative images shown in **g** (scale bar, 50 μ m). **h**, Sleep profiles over a 48-hour period for flies with EEC-specific *upd3* knockdown under a transition from standard to 1% agar starvation diet compared to control flies (N=27-32). **i and j**, Daytime and nighttime sleep durations for flies with EEC-specific *upd3* knockdown under standard diet and 1% agar starvation diet conditions (N=27-32). Statistical tests: *t*-tests for panel **f**; Kruskal-Wallis ANOVA with Dunn's multiple comparisons was used for panels **b, c, d, and e**; two-way ANOVA with Sidak's multiple comparisons was used for panels **i and j**. Data are presented as mean $\pm$ SEM. ns, non-significant ($p > 0.05$).

Figure S4.

Expression of *upd2* and *upd3*.

Transcript levels of **a**, *upd2*, and **b**, *upd3* in heads of animals with RNAi-mediated knockdown in EECs using *voilà-GAL4* in combination with *Tubulin-GAL80^{TS}* (*voilà>*) (N=6). Statistical analyses were performed using *t*-tests for panel a and b. Data are presented as mean $\pm$ SEM.

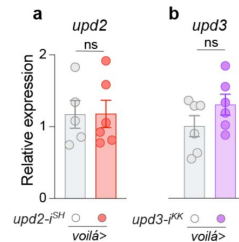


Figure S5.

Impact of glia-specific *dome* knockdown on sleep patterns following oxidative stress and after sleep deprivation.

a-c, Daytime sleep measured over three days, starting with a normal diet, followed by a day with 1% H_2O_2 -supplemented food to induce oxidative stress, and concluding with a return to a normal diet to assess recovery, in flies with UAS-RNAi constructs without *dome* knockdown (UAS-RNAi controls) and those with glia-specific *dome* knockdown (N=25-31). **d**, Sleep profiles during a night that included a 6-hour period of sleep deprivation, followed by a recovery phase, in control flies and those with glia-specific *dome* knockdown (N=17-24). **e**, Sleep quantity measured during the first 2 hours of the recovery period following sleep deprivation in control flies and those with glia-specific *dome* knockdown (N=17-24). Statistical tests: Kruskal-Wallis ANOVA with Dunn's multiple comparisons for panels a, b, and e; ordinary one-way ANOVA with Dunnett's multiple comparisons for panel c. Data are presented as mean $\pm$ SEM. ns, non-significant ($p > 0.05$)

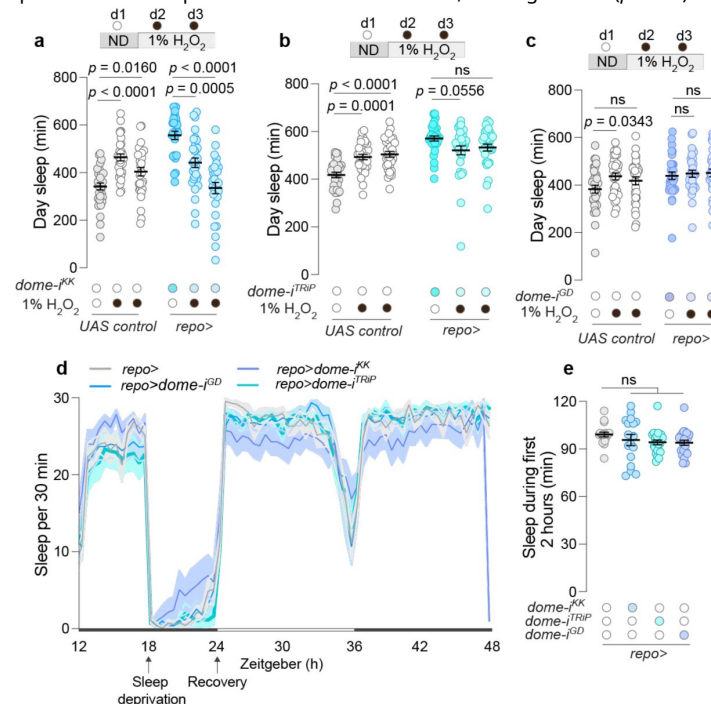


Figure S6.

AstA signaling from EECs promotes wakefulness.

a, *AstA-R2* expression in heads of flies with knockdown of *AstA-R2* in BBB glia using *moody-GAL4* (*moody*>) compared to the control group (N=5). **b**, Sleep profiles over two consecutive days, encompassing one day under standard diet conditions followed by one day on oxidative stress conditions, induced by 1% H₂O₂-containing food, in animals with BBB glia-specific knockdown of *AstA-R1* or *AstA-R2* and controls (N=30-32). **c**, Daytime sleep durations over two consecutive days, encompassing one day under standard diet conditions followed by one day on oxidative stress conditions, induced by 1% H₂O₂-containing food, in animals with BBB glia-specific knockdown of *AstA-R1* or *AstA-R2* and controls (N=30-32). **d**, Sleep profiles, and **e**, daytime sleep on standard food in animals with *AstA* knockdown in *AstA*⁺ EECs and *UAS-RNAi* controls (N=30-31). **f**, Single-cell RNA sequencing data from adult *Drosophila* midguts from the Fly Cell Atlas dataset⁵² were visualized with the SCoPe viewer. *AstA* (red), *Tk* (blue), and *TrpA1* (green) expression is shown across the EEC clusters. *AstA*-positive EECs form a distinct cluster that does not overlap with *TrpA1* expression. In contrast, *Tk*-positive EECs form a separate cluster, and a subset of these cells express *TrpA1*. Right: Zoom-in view of the *Tk*-positive EEC cluster highlights cells co-expressing *Tk* and *TrpA1*. These data indicate that *TrpA1* is selectively expressed in *Tk*-positive EECs and not in *AstA*-positive EECs. Statistical tests used: Two-sided unpaired *t*-tests for panel a and e; two-way ANOVA with Sidak's multiple comparisons was used for panel c. Data are presented as mean ± SEM.

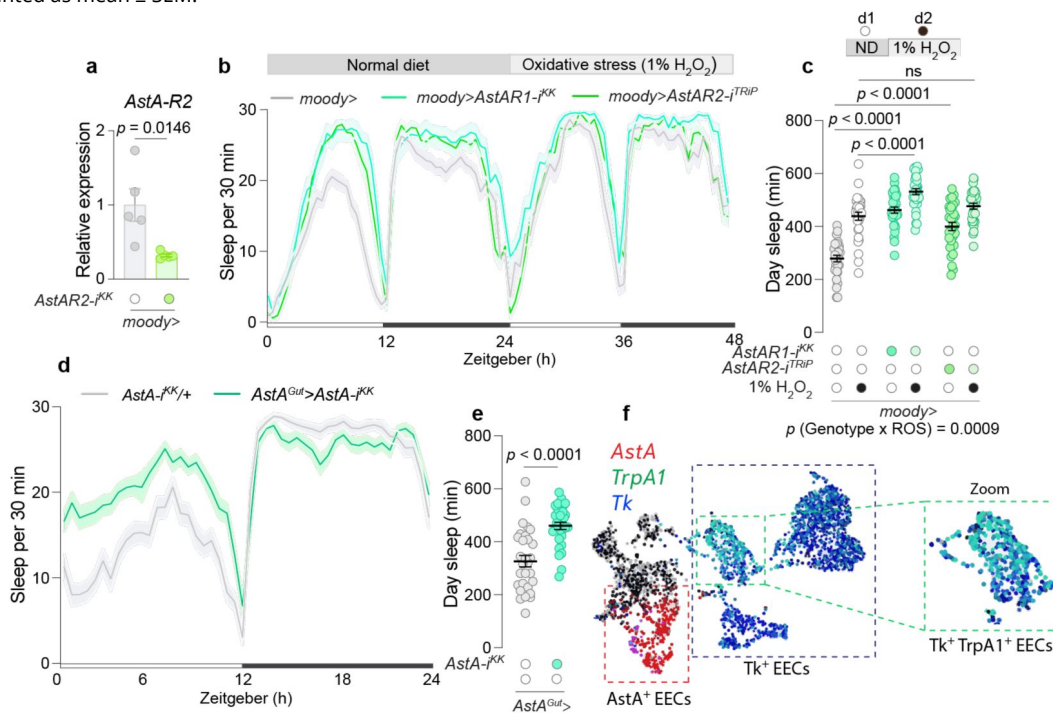


Table S1

Oligos used for cloning the *upd2* and *upd3* CRISPR constructs, with gRNA sequences indicated in bold and underlined.

Oligo	Sequence
<i>upd2fwd</i>	CGGCCCGGGTTCGATTCCCGCCGATGCA <u>TTCTCGCCCGCTCGATTGGT</u> GTTTCAGAGCTATGCTGGAAAC
<i>upd2rev</i>	ATTTTAAGTTCGCTATTTCTAGCTCTAAAC <u>CATGCAACAGTCACTGACGA</u> TGCACCAGCCGGGAATCGAACC
<i>upd3fwd</i>	CGGCCCGGGTTCGATTCCCGCCGATGCA <u>GACAACGTAACGTAACCGAC</u> GTTTCAGAGCTATGCTGGAAAC
<i>upd3rev</i>	ATTTTAAGTTCGCTATTTCTAGCTCTAAAC <u>TTTGGTCTGTAGATTCTGC</u> TGCACCAGCCGGGAATCGAACC

Gene	Primer	Sequence
<i>upd2</i>	Forward	CGGAACATCACGATGAGCGAA
<i>upd2</i>	Reverse	TCGGCAGGAACTTGTACTCG
<i>upd3</i>	Forward	TGTCGAGAAGAACAAGTGGCG
<i>upd3</i>	Reverse	CGTGGCGAAGGTTCAACTGT
<i>Dome</i>	Forward	CTCACGTCTCGACTGGGAAC
<i>Dome</i>	Reverse	AGAATGGTGCTTGTGAGGCA
<i>AstA</i>	Forward	CGCCTGCCGGTCTATAACTT
<i>AstA</i>	Reverse	CTTGTTCTGTGCGCCAGGTC
<i>AstAR1</i>	Forward	GCCACTGGAAACGGTAGTATC
<i>AstAR1</i>	Reverse	CGTGTGTTCCGAGGTGAATG
<i>AstAR2</i>	Forward	CGCAGTGTCCAGTACCTGATT
<i>AstAR2</i>	Reverse	GAGCGAATGGGATGAACCAC
<i>Rp49</i>	Forward	AGTATCTGATGCCCAACATCG
<i>Rp49</i>	Reverse	CAATCTCCTTGCGCTTCTTG

Table S1

List of qPCR primers

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Joint Public Review:

Summary:

Malita and colleagues investigated the mechanism by which infections increase sleep in *Drosophila*. Their work is important because it further supports the idea that the blood brain barrier is involved in brain-body communication, and because it advances the field of sleep research. Using knock-down and knock-out of cytokines and cytokine receptors specifically in the endocrine cells of the gut (cytokines) as well as in the glia forming the blood-brain barrier (BBB) (cytokines receptors), the authors show that cytokines, upd2 and upd3, secreted by entero-endocrine cells in response to infections increase sleep through the Dome receptor in the BBB. They also show that gut-derived Allatostatin (Alst) A promotes wakefulness by

inhibiting the Alst A signaling that is mediated by Alst receptors expressed in BBB glia. Their results suggest there may be additional mechanisms that promote elevated sleep during gut inflammation. The evidence supporting most of their claims is compelling. Nevertheless, the activation of the sleep-promoting pathway by infection should be accomplished through bacterial infection of the gut.

Strengths:

The work is, in general, supported by well-designed and well-performed experiments, especially those that show that the endocrine cells from the gut are the sources of the Upd cytokines, the effects of these cytokines on daytime sleep, and that the glial cells of the BBB are the target cell for the Upds action. In addition, the evidence associating the downregulation of Alst receptors in the BBB by Upd and Jak/Stat pathways is compelling.

Weaknesses:

(1) The model of gut inflammation that is used is based on the increase in reactive oxygen species (ROS) that is caused by adding 1% H₂O₂ to the food. The use of the model is supported rather weakly by two papers (ref. 26 and 27). The paper by Jiang et al. (26) shows that the infection by *Pseudomonas entomophila* induces cytokine responses Upd2 and 3, which are also induced by the Jnk pathway; there is no mention of ROS. Buchon et al. (27) is a review that refers to results that indicate that as part of the immune response to pathogens in the gut, there is production of ROS by the NADPH oxidase DUOX. Thus, there is no strong support for the use of this model.

(2) There is no support for the use of ROS in the food instead a direct infection by pathogenic bacteria. It is known that ROS causes damage in the gut epithelium, which in turn induces the expression of the cytokines studied, which might be independent of infection and confound the results.

<https://doi.org/10.7554/eLife.99999.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Joint Public Review:

Summary:

The authors sought to elucidate the mechanism by which infections increase sleep in Drosophila. Their work is important because it further supports the idea that the blood-brain barrier is involved in brain-body communication, and because it advances the field of sleep research. Using knock-down and knock-out of cytokines and cytokine receptors specifically in the endocrine cells of the gut (cytokines) as well as in the glia forming the blood-brain barrier (BBB) (cytokines receptors), the authors show that cytokines, upd2 and upd3, secreted by entero-endocrine cells in response to infections increase sleep through the Dome receptor in the BBB. They also show that gut-derived Allatostatin (Alst) A promotes wakefulness by inhibiting Alst A signaling that is mediated by Alst receptors expressed in BBB glia. Their results suggest there may be additional mechanisms that promote elevated sleep during gut inflammation.

The authors suggest that upd3 is more critical than upd2, which is not sufficiently addressed or explained. In addition, the study uses the gut's response to reactive oxygen molecules as a proxy for infection, which is not sufficiently justified. Finally, further

verification of some fundamental tools used in this paper would further solidify these findings making them more convincing.

Strengths:

(1) The work addresses an important topic and proposes an intriguing mechanism that involves several interconnected tissues. The authors place their research in the appropriate context and reference related work, such as literature about sickness-induced sleep, ROS, the effect of nutritional deprivation on sleep, sleep deprivation and sleep rebound, upregulated receptor expression as a compensatory mechanism in response to low levels of a ligand, and information about *Alst A*.

(2) The work is, in general, supported by well-performed experiments that use a variety of different tools, including multiple RNAi lines, CRISPR, and mutants, to dissect both signal-sending and receiving sides of the signaling pathway.

(3) The authors provide compelling evidence that shows that endocrine cells from the gut are the source of the *upd* cytokines that increase daytime sleep, that the glial cells of the BBB are the targets of these *upds*, and that *upd* action causes the downregulation of *Alst* receptors in the BBB via the *Jak/Stat* pathways.

We are pleased that the reviewers recognized the strength and significance of our findings describing a gut-to-brain cytokine signaling mechanism involving the blood-brain barrier (BBB) and its role in regulating sleep, and we thank them for their comments.

Weaknesses:

(1) There is a limited characterization of cell types in the midgut which are classically associated with *upd* cytokine production.

We thank the reviewer for raising this point. Although several midgut cell types (including the absorptive enterocytes) may indeed produce Unpaired (Upd) cytokines, our study specifically focused on enteroendocrine cells (EECs), which are well-characterized as secretory endocrine cells capable of exerting systemic effects. As detailed in our response to Results point #2 (please see below), we show that EEC-specific manipulation of Upd signaling is both necessary and sufficient to regulate sleep in response to intestinal oxidative stress. These findings support the role of EECs as a primary source of gut-derived cytokine signaling to the brain. To acknowledge the possible involvement of other source, we have also added a statement to the Discussion in the revised manuscript noting that other, non-endocrine gut cell types may contribute to systemic Unpaired signaling that modulates sleep.

(2) Some of the main tools used in this manuscript to manipulate the gut while not influencing the brain (e.g., *Voilà* and *Voilà + R57C10-GAL80*), are not directly shown to not affect gene expression in the brain. This is critical for a manuscript delving into intra-organ communication, as even limited expression in the brain may lead to wrong conclusions.

We agree with the reviewer that this is an important point. To address it, we performed additional validation experiments to assess whether the *voilà-GAL4* driver in combination with *R57C10-GAL80* (*EEC>*) influences *upd2* or *upd3* expression in the brain. Our results show that manipulation using *EEC>* alters *upd2* and *upd3* expression in the gut (Fig. 1a,b), with new data showing that this does not affect their expression levels in neuronal tissues (Fig. S1a), supporting the specificity of our approach. These new data are now included in the revised manuscript and described in the Results section. This additional validation strengthens our conclusion that the observed sleep phenotypes result from gut-specific cytokine signaling, rather than from effects on Unpaired cytokines produced in the brain.

(1) >(3) The model of gut inflammation used by the authors is based on the increase in reactive oxygen species (ROS) obtained by feeding flies food containing 1% H₂O₂. The use of this model is supported by the authors rather weakly in two papers (refs. 26 and 27): The paper by Jiang et al. (ref. 26) shows that the infection by *Pseudomonas entomophila* induces cytokine responses *upd2* and *3*, which are also induced by the Jnk pathway. In addition, no mention of ROS could be found in Buchon et al. (ref 27); this is a review that refers to results showing that ROS are produced by the NADPH oxidase DUOX as part of the immune response to pathogens in the gut. Thus, there is no strong support for the use of this model.

We thank the reviewer for raising this point. We agree that the references originally cited did not sufficiently justify the use of H₂O₂ feeding as a model of gut inflammation. To address this, we have revised the Results section to clarify that we use H₂O₂ feeding as a controlled method to elevate intestinal ROS levels, rather than as a general model of inflammation. This approach allows us to investigate the specific effects of ROS-induced cytokine signaling in the gut. We have also added additional citations to support the physiological relevance of this model. For instance, Tamamouna et al. (2021) demonstrated that H₂O₂ feeding induces intestinal stem-cell proliferation – a response also observed during bacterial infection – and Jiang et al. (2009) showed that enteric infections increase *upd2* and *upd3* expression, which we similarly observe following H₂O₂ feeding (Fig. 3a). These findings support the use of H₂O₂ as a tool to mimic specific ROS-linked responses in the gut. We believe this targeted and tractable model is a strength of our study, enabling us to dissect how intestinal ROS modulates systemic physiology through cytokine signaling

Additionally, we have included a statement in the Discussion acknowledging that ROS generated during infection may activate signaling mechanisms distinct from those triggered by chemically induced oxidative stress, and that exploring these differences in future studies may yield important insights into gut–brain communication. These revisions provide a stronger justification for our model while more accurately conveying both its relevance and its limitations.

(2) >(4) Likewise, there is no support for the use of ROS in the food instead a direct infection by pathogenic bacteria. Furthermore, it is known that ROS damages the gut epithelium, which in turn induces the expression of the cytokines studied. Thus the effects observed may not reflect the response to infection. In addition, Majcin Dorcikova et al. (2023). Circadian clock disruption promotes the degeneration of dopaminergic neurons in male *Drosophila*. Nat Commun. 2023 14(1):5908. doi: 10.1038/s41467-02341540-y report that the feeding of adult flies with H₂O₂ results in neurodegeneration if associated with circadian clock defects. Thus, it would be important to discuss or present controls that show that the feeding of H₂O₂ does not cause neuronal damage.

We thank the reviewer for this thoughtful follow-up point. We would like to clarify that we do not claim that the effects observed in our study directly reflect the full response to enteric infection. As outlined in our revised response to comment 3, we have updated the manuscript to more precisely describe the H₂O₂-feeding paradigm as a model that induces local intestinal ROS responses comparable to, but not equivalent to, those observed during pathogenic challenges. This revised framing highlights both the potential similarities and differences between chemically induced oxidative stress and infection-induced responses. Indeed, in the revised Discussion, we now explicitly acknowledge that ROS generated during infection may engage distinct signaling mechanisms compared to exogenous H₂O₂ and emphasize the value of future studies in delineating these pathways. We are currently pursuing this direction in an independent ongoing study investigating the effects of enteric infections. However, for the present work, we chose to focus on the effects of ROS-induced responses in isolation, as this provides a clean and well-controlled context to dissect the specific contribution of oxidative stress to cytokine signaling and sleep regulation.

To further address the reviewer's concern, we have also included new data (a TUNEL stain for apoptotic DNA fragmentation) in the revised manuscript showing that H₂O₂ feeding does not damage neuronal tissues under our experimental conditions (Fig. S3f,g). This addresses the point raised regarding the potential neurotoxicity of H₂O₂, as described by Majcin Dorcikova et al. (2023), and supports the specificity of the sleep phenotypes observed in our study. We believe these revisions and clarifications strengthen the manuscript and make our interpretation more precise.

(3) >(5) The novelty of the work is difficult to evaluate because of the numerous publications on sleep in *Drosophila*. Thus, it would be very helpful to read from the authors how this work is different and novel from other closely related works such as: Li et al. (2023) Gut AstA mediates sleep deprivation-induced energy wasting in *Drosophila*. *Cell Discov.* 23;9(1):49. doi: 10.1038/s41421-023-00541-3.

Our work highlights a distinct role for gut-derived AstA in sleep regulation compared to findings by Lin et al. (*Cell Discovery*, 2023)[1], who showed that gut AstA mediates energy wasting during sleep deprivation. Their study focused on the metabolic consequences of sleep loss, proposing that sleep deprivation increases ROS in the gut, which then promotes the release of the glucagon-like hormone adipokinetic hormone (AKH) through gut AstA signaling, thereby triggering energy expenditure.

In contrast, our study addresses the inverse question – how ROS in the gut influences sleep. In our model, intestinal ROS promotes sleep, raising the intriguing possibility – cleverly pointed out by the reviewers – that ROS generated during sleep deprivation might promote sleep by inducing Unpaired cytokine signaling in the gut. According to our findings, this suppresses wake-promoting AstA signaling in the BBB, providing a mechanism to promote sleep as a restorative response to gut-derived oxidative stress and potentially limiting further ROS accumulation. Importantly, our findings support a wakepromoting role for EEC-derived AstA, demonstrated by several lines of evidence. First, EEC-specific knockdown of AstA increases sleep. Second, activation of AstA⁺ EECs using the heat-sensitive cation channel Transient Receptor Potential A1 (TrpA1) reduces sleep, and this effect is abolished by simultaneous knockdown of *AstA*, indicating that the sleep-suppressing effect is mediated by AstA and not by other peptides or secreted factors released by these cells. Third, downregulation of *AstA* receptor expression in BBB glial cells increases sleep, further supporting the existence of a functional gut AstA–glia arousal pathway. We have now included new data in the revised manuscript showing that AstA release from EECs is downregulated during intestinal oxidative stress (Fig. 7k,l,m). This suggests that this wake-promoting signal is suppressed both at its source (the gut endocrine cells), by unknown means, and at its target, the BBB, via Unpaired cytokine signaling that downregulates AstA receptor expression. This coordinated downregulation may serve to efficiently silence this arousal-promoting pathway and facilitate sleep during intestinal stress. These new data, along with an expanded discussion, provide further mechanistic insight into gut-derived AstA signaling and strengthen our proposed model.

This contrasts with the interpretation by Lin et al., who observed increased AstA peptide levels in EECs after antioxidant treatment and interpreted this as peptide retention. However, peptide accumulation may result from either increased production or decreased release, and peptide levels alone are insufficient to distinguish between these possibilities. To resolve this, we examined *AstA* transcript levels, which can serve as a proxy for production. Following oxidative stress (24 h of 1% H₂O₂ feeding and the following day), when animals show increased sleep (Fig. 7e), we observed a decrease in *AstA* transcript levels followed by an increase in peptide levels (Fig. 7k,l,m), suggesting that oxidative stress leads to reduced gut AstA production and release. Furthermore, we recently found that a class of EECs that produce the hormone Tachykinin (Tk) and are distinct from the AstA⁺ EECs express the ROSsensitive cation channel *TrpA1* (Ahrentlöv et al., 2025, *Nature Metabolism*2). In these Tk⁺

EECs, TrpA1 mediates ROS-induced Tk hormone release. In contrast, single-cell RNA-seq data[3] do not support *TrpA1* expression in AstA⁺ EECs, consistent with our findings that ROS does not promote AstA release – an effect that would be expected if TrpA1 were functionally expressed in AstA⁺ EECs. This contradicts the findings of Lin et al., who reported *TrpA1* expression in AstA⁺ EECs. We have now included relevant single-cell data in the revised manuscript (Fig. S6f) showing that *TrpA1* is specifically expressed in Tk⁺ EECs, but not in AstA⁺ EECs, and we have expanded the discussion to address discrepancies in TrpA1 expression and AstA regulation.

Taken together, our results reveal a dual-site regulatory mechanism in which Unpaired cytokines released from the gut act at the BBB to downregulate AstA receptor expression, while AstA release from EECs is simultaneously suppressed. We thank the reviewers for raising this important point. We have also included a discussion the other point raised by the reviewers – the possibility that ROS generated during sleep deprivation may engage the same signaling pathways described here, providing a mechanistic link between sleep deprivation, intestinal stress, and sleep regulation.

Recommendations for the authors:

A- Material and Methods:

(1) Feeding Assay: The cited publication (doi.org:10.1371/journal.pone.0006063) states: "For the amount of label in the fly to reflect feeding, measurements must therefore be confined to the time period before label egestion commences, about 40 minutes in Drosophila, a time period during which disturbance of the flies affects their feeding behavior. There is thus a requirement for a method of measuring feeding in undisturbed conditions." Was blue fecal matter already present on the tube when flies were homogenized at 1 hour? If so, the assay may reflect gut capacity rather than food passage (as a proxy for food intake). In addition, was the variability of food intake among flies in the same tube tested (to make sure that 1-2 flies are a good proxy for the whole population)?

We agree that this is an important point for feeding experiments. We are aware of the methodological considerations highlighted in the cited study and have extensive experience using a range of feeding assays in *Drosophila*, including both short- and long-term consumption assays (e.g., dye-based and CAFE assays), as well as automated platforms such as FLIC and FlyPAD (Nature Communications, 2022; Nature Metabolism, 2022; and Nature Metabolism, 2025)[2,4,5].

For the dye-based assay, we carefully selected a 1-hour feeding window based on prior optimization. Since animals were not starved prior to the assay, shorter time points (e.g., 30 minutes) typically result in insufficient ingestion for reliable quantification. A 1-hour period provides a robust readout while remaining within the timeframe before significant label excretion occurs under our experimental conditions. To support the robustness of our findings, we complemented the dye-based assay with data from FLIC, which enables automated, high-resolution monitoring of feeding behavior in undisturbed animals over extended periods. The FLIC results were consistent with the dye-based data, strengthening our confidence in the conclusions. To minimize variability and ensure consistency across experiments, all feeding assays were performed at the same circadian time – Zeitgeber Time 0 (ZT0), corresponding to 10:00 AM when lights are turned on in our incubators. This time point coincides with the animals' natural morning feeding peak, allowing for reproducible comparisons across conditions. Regarding variability among flies within tubes, each biological replicate in the dye assay consisted of 1–2 flies, and results were averaged across multiple replicates. We observed good consistency across samples, suggesting that these small groups reliably reflect group-level feeding behavior under our conditions.

(2) Biological replicates: whereas the number of samples is clearly reported in each figure, the number of biological replicates is not indicated. Please include this information either in Material and methods or in the relevant figure legends. Please also include a description of what was considered a biological replicate.

We have now clarified in the Materials and Methods section under Statistics that all replicates represent independent biological samples, as suggested by the reviewers.

(3) Control Lines: please indicate which control lines were used instead of citing another publication. If preferred, this information could be supplied as a supplementary table.

We now provide a clear description of the control lines used in the Materials and Methods section. Specifically, all GAL4 and GAL80 lines used in this study were backcrossed for several generations into a shared w^{1118} background and then crossed to the same w^{1118} strain used as the genetic background for the *UAS-RNAi*, *crispr*, or overexpression lines. This approach ensures, to a strong approximation, that the only difference between control and experimental animals is the presence or absence of the UAS transgene.

(4) Statistical analyses: for some results (e.g., those shown in Figure 3d), it could be useful to test the interaction between genotype and treatment.

We thank the reviewer for this helpful suggestion. In response, we have now performed two-way ANOVA analyses to assess genotype $\times$ treatment (diet) interaction effects for the relevant data, including those shown in Figure 3d as well as additional panels where animals were exposed to oxidative stress and sleep phenotypes were measured. We have added the corresponding interaction p-values in the updated figure legends for Figures 3d, 3k, 5a–c, 5f, 5h, 5i, 6c, 6e, and 7e. All of these tests revealed significant interaction effects, supporting the conclusion that the observed differences in sleep phenotypes are specifically dependent on the interaction between genetic manipulation (e.g., cytokine or receptor knockdown) and oxidative stress. These additions reinforce the interpretation that Unpaired cytokine signaling, glial JAK-STAT pathway activity, and AstA receptor regulation functionally interact with intestinal ROS exposure to modulate sleep. We thank the reviewer for suggesting this improvement.

(5) Reporting of p values. Some are reported as specific values whereas others are reported as less than a specific value. Please make this reporting consistent across different figures.

All p-values reported in the manuscript are exact, except in cases where values fall below $p < 0.0001$. In those instances, we use the inequality because the Prism software package (GraphPad, version 10), which was used for all statistical analyses, does not report more precise values. We believe this reporting approach reflects standard practice in the field.

(6) Please include the color code used in each figure, either in the figure itself or in the legend.

We have now clarified the color coding in all relevant figures. In particular, we acknowledge that the meaning of the half-colored circles used to indicate H_2O_2 treatment was not previously explained. These have now been clearly labeled in each figure to indicate treatment conditions.

(7) The scheme describing the experimental conditions and the associated chart is confusing. Please improve.

We have improved the schematic by replacing “ROS” with “H₂O₂” to more clearly indicate the experimental condition used. Additionally, we have added the corresponding circle annotations so that they now also appear consistently above the relevant charts. This revised layout enhances clarity and helps readers more easily interpret the experimental conditions. We believe these changes address the reviewer’s concern and make the figure significantly more intuitive.

1. Please indicate which line was used for *upd-Gal4* and the evidence that it faithfully reflects *upd3* expression.

We have now clarified in the Materials and Methods section that the *upd3-GAL4* line used in our study is Bloomington stock #98420, which drives *GAL4* expression under the control of approximately 2 kb of sequence upstream of the *upd3* start codon. This line has previously been used as a transcriptional reporter for *upd3* activity. The only use of this line was to illustrate reporter expression in the EECs. To support this aspect of *Upd3* expression, we now include new data in the revised manuscript using fluorescent in situ hybridization (FISH) against *upd3*, which confirms the presence of *upd3* transcripts in *prospero*-positive EECs of the adult midgut (Fig. S1b). Additionally, we show that *upd3* transcript levels are significantly reduced in dissected midguts following EEC-specific knockdown using multiple independent RNAi lines driven by *voilà-GAL4*, both alone and in combination with *R57C10-GAL80*, consistent with endogenous expression in these cells (Fig. 1a,b).

To further address the reviewer’s concern and provide additional support for the endogenous expression of *upd3* in EECs, we performed targeted knockdown experiments focusing on molecularly defined EEC subpopulations. The adult *Drosophila* midgut contains two major EEC subtypes characterized by their expression of Allatostatin C (AstC) or Tachykinin (Tk), which together encompass the vast majority of EECs. To selectively manipulate these populations, we used *AstC-GAL4* and *Tk-GAL4* drivers – both knock-in lines in which *GAL4* is inserted at the respective endogenous hormone loci. This design enables precise *GAL4* expression in AstC- or Tk-expressing EECs based on their native transcriptional profile. To eliminate confounding neuronal expression, we combined these drivers with *R57C10GAL80*, restricting *GAL4* activity to the gut and generating *AstC^{Gut}* and *Tk^{Gut}* drivers. Using these tools, we knocked down *upd2* and *upd3* selectively in the AstC- or Tk-positive EECs. Knockdown of either cytokine in AstC-positive EECs significantly increased sleep under homeostatic conditions, recapitulating the phenotype observed with knockdown in all EECs (Fig. 1m-o). In contrast, knockdown of *upd2* or *upd3* in Tk-positive EECs had no effect on sleep (Fig. 1p-r). Furthermore, we show in the revised manuscript that selective knockdown of *upd2* or *upd3* in AstC-positive EECs abolishes the H₂O₂-induced increase in sleep (Fig. 3f-h). These findings demonstrate that Unpaired cytokine signaling from AstC-positive EECs is essential for mediating the sleep response to intestinal oxidative stress, highlighting this specific EEC subtype as a key source of cytokine-driven regulation in this context. These new results indicate that AstC-positive EECs are a primary source of the Unpaired cytokines that regulate sleep, while Tk-positive EECs do not appear to contribute to this function. Importantly, *upd3* transcript levels were significantly reduced in dissected midguts following *AstC^{Gut}* driven knockdown (Fig. S1r), further confirming that *upd3* is endogenously expressed in AstC-positive EECs. Thus we have bolstered our confidence that *upd3* is indeed expressed in EECs, as illustrated by the reporter line, through several means.

(9) Please indicate which GFP line was used with *upd-Gal4* (CD8, NLS, un-tagged, etc). The Material and Methods section states that it was “*UAS-mCD8::GFP* (#5137);”, however, the stain does not seem to match a cell membrane pattern but rather a nuclear or cytoplasmic pattern. This information would help the interpretation of Figure 1C.

We confirm that the GFP reporter line used with *upd3*-GAL4 was obtained from Bloomington stock #98420. As noted by the Bloomington *Drosophila* Stock Center, “the identity of the UAS-GFP transgene is a guess,” and the subcellular localization of the GFP fusion is therefore uncertain. We agree with the reviewer that the signal observed in Figure 1c does not display clear membrane localization and instead appears diffuse, consistent with cytoplasmic or partially nuclear localization. In any case, what we find most salient is the reporter’s labeling of Prospero-positive EECs in the adult midgut, consistent with *upd3* expression in these cells. This conclusion is further supported by multiple lines of evidence presented in the revised manuscript, as mentioned above in response to question #8: (1) fluorescent in situ hybridization (FISH) for *upd3* confirms expression in EECs (Fig. S1b), (2) EEC-specific RNAi knockdown of *upd3* reduces transcript levels in dissected midguts, and (3) publicly available single-cell RNA sequencing datasets[3] also indicate that *upd3* is expressed at low levels in a subset of adult midgut EECs under normal conditions. We have also clarified in the revised Materials and Methods section that GFP localization is undefined in the *upd3*-GAL4 line, to guide interpretation of the reporter signal.

B- Results

(1) Figure 1: According to previous work (10.1016/j.celrep.2015.06.009, <http://flygutseq.buchonlab.com/data?gene=upd3%0D%0A>), in basal conditions *upd3* is expressed as following: ISC (35 RPKM), EB (98 RPKM), EC (57 RPKM), and EEC (8 RPKM). Accordingly, even complete KO in EECs should eliminate only a small fraction of *upd3* from whole guts, even less considering the greater abundance of other cell types such as ECs compared to EECs. It would be useful to understand where this discrepancy comes from, in case it is affecting the conclusion of the manuscript. While this point per se does not affect the main conclusions of the manuscript, it makes the interpretation of the results more difficult.

We acknowledge the previously reported low expression of *upd3* in EECs. However, the FlyGut-seq site appears to be no longer available, so we could not directly compare other related genes. Nonetheless, our data – based on in situ hybridization, reporter expression, and multiple RNAi knockdowns – consistently support *upd3* expression in EECs. These complementary approaches strengthen the conclusion that EECs are an important source of systemic *upd3* under the conditions tested.

(2) Figure 1: The *upd2-3* mutants show sleep defects very similar to those of *EEC>RNAi* and *>Cas9*. It would thus be helpful to try to KO *upd3* with other midgut drivers (An EC driver like *Myo1A* or *5966GS* and a progenitor driver like *Esg* or *5961GS*) to validate these results. Such experiments might identify precisely which cells are involved in the gut-brain signaling reported here.

We appreciate the reviewer’s suggestion and agree that exploring other potential sources of Upd3 in the gut is an interesting direction. In this study, we have focused on EECs, which are the primary hormone-secreting cells in the intestine and thus the most likely candidates for mediating systemic effects such as gut-to-brain signaling. While it is possible that other gut cell types – such as enterocytes (e.g., *Myo1A*⁺) or intestinal progenitors (e.g., *Esg*⁺) – also contribute to Upd3 production, these cells are not typically endocrine in nature. Demonstrating their involvement in gutto-brain communication would therefore require additional, extensive validation beyond the scope of the current study. Importantly, our data show that manipulating Upd3 specifically in EECs is both necessary and sufficient to modulate sleep in response to intestinal ROS, strongly supporting the conclusion that EEC-derived cytokine signaling underlies the observed phenotype. In contrast, manipulating cytokines in other gut cells could produce indirect effects – such as altered proliferation, epithelial integrity, or immune responses – that complicate the interpretation of behavioral

outcomes like sleep. For these reasons, we chose to focus on EECs as the source of endocrine signals mediating gut-to-brain communication. However, to address this point raised by the reviewer, we have now included a statement in the Discussion acknowledging that other non-endocrine gut cell types may also contribute to the systemic Unpaired signaling that modulates sleep in response to intestinal oxidative stress.

(3) Figure 3: "This effect mirrored the upregulation observed with EEC-specific overexpression of *upd3*, indicating that it reflects physiologically relevant production of *upd3* by the gut in response to oxidative stress." Please add (Figure 3a) at the end of this sentence.

We have now added "(Figure 3a)" at the end of the sentence to clearly reference the relevant data.

(4) For Figure 3b, do you have data showing that the increased amount of sleep was due to the addition of H_2O_2 per se, rather than the procedure of adding it?

We have added new data to address this point. To ensure that the observed sleep increase was specifically due to the presence of H_2O_2 and not an effect of the food replacement procedure, we performed a control experiment in which animals were fed standard food prepared using the same protocol and replaced daily, but without H_2O_2 . These animals did not exhibit increased sleep, confirming that the sleep effect is attributable to intestinal ROS rather than the supplementation procedure itself (Fig. S3a). Thanks for the suggestion.

(5) In the text it is stated that "Since 1% H_2O_2 feeding induced robust responses both in *upd3* expression and in sleep behavior, we asked whether gut-derived Unpaired signaling might be essential for the observed ROS-induced sleep modulation. Indeed, EEC-specific RNAi targeting *upd2* or *upd3* abolished the sleep response to 1% H_2O_2 feeding." While it is indeed true that there is no additional increase in sleep time due to EEC>*upd3* RNAi, it is also true that EEC>*upd3* RNAi flies, without any treatment, have already increased their sleep in the first place. It is then possible that rather than unpaired signaling being essential, an upper threshold for maximum sleep allowed by manipulation of these processes was reached. It would be useful to discuss this point.

Several findings argue against a ceiling effect and instead support a requirement for Unpaired signaling in mediating ROS-induced sleep. Animals with EEC-specific *upd2* or *upd3* knockdown or null mutation not only fail to increase sleep following H_2O_2 treatment but actually exhibit reduced sleep during oxidative stress (Fig. 3e, k, l; Fig. 5e, f), suggesting that Unpaired signaling is required to sustain sleep under these conditions. Similarly, animals with glial dome knockdown also show reduced sleep under oxidative stress, closely mirroring the phenotype of EEC-specific *upd3* RNAi animals (Fig. 5a–c, g–i). These results support the conclusion that gut-to-glia Unpaired cytokine signaling is necessary for maintaining elevated sleep during oxidative stress. In the absence of this signaling, animals exhibit increased wakefulness. We identify AstA as one such wake-promoting signal that is suppressed during intestinal stress. We present new data showing that this pathway is downregulated not only via Unpaired-JAK/STAT signaling in glial cells but also through reduced AstA release from the gut in the revised manuscript. This model, in which Unpaired cytokines promote sleep during intestinal stress by suppressing arousal pathways, is discussed throughout the manuscript to address the reviewer's point.

(6) In Figure 3k, the dots highlighting the experiment show an empty profile, a full one, and a half one. Please define what the half dots represent.

We have now clarified the color coding in all relevant figures. Specifically, we acknowledge that the meaning of the half-colored circles indicating H_2O_2 treatment was not previously defined – it indicates washout or recovery time. In the revised version, these symbols are now clearly labeled in each figure to indicate the treatment condition, ensuring consistent and intuitive interpretation across all panels.

(7) The authors used appropriate GAL4 and RNAi lines to the knockdown dome, a upd2/3 JAK-STATlinked receptor, specifically in neurons and glia, respectively, in order to identify the CNS targets of upd2/3 cytokines produced by enteroendocrine cells (EECs). Pan-neuronal dome knockdown did not alter daytime sleep in adult females, yet pan-glial dome knockdown phenocopied effects of upd2/3 knockdown in EECs. They also observed that EEC-specific knockdown of upd2 and upd3 led to a decrease in JAK-STAT reporter activity in repo-positive glial cells. This supports the authors' conclusion that glial cells, not neurons, are the targets by which unpaired cytokines regulate sleep via JAK-STAT signaling. However, they do not show nighttime sleep data of pan-neuronal and pan-glial dome knockdowns. It would strengthen their conclusion if the nighttime sleep of pan-glial dome knockdown phenocopied the upd2/3 knockdowns as well, provided the pan-neuronal dome knockdown did not alter nighttime sleep.

We have now added nighttime sleep data for both pan-glial and pan-neuronal *domeless* knockdowns in the revised manuscript (Fig. 2a). Glial knockdown increased nighttime sleep, similar to EEC-specific *upd2/3* knockdown, while neuronal knockdown had no effect. These results further support the glial cells' being the relevant target of gut-derived Unpaired signaling.

(8) The authors only used one method to induce oxidative stress (hydrogen peroxide feeding). It would strengthen their argument to test multiple methods of inducing oxidative stress, such as lipopolysaccharide (LPS) feeding. In addition, it would be useful to use a direct bacterial infection to confirm that in flies, the infection promotes sleep. Additionally, flies deficient in Dome in the BBB and infected should not be affected in their sleep by the infection. These experiments would provide direct support for the mechanism proposed. Finally, the authors should add a primary reference for using ROS as a model of bacterial infection and justify their choice better.

We agree that directly comparing different models of intestinal stress, such as bacterial infection or LPS feeding, would provide valuable insight into how gut-derived signals influence sleep in response to infection. As noted in our detailed responses above, we now include an expanded rationale for our use of H_2O_2 feeding as a controlled and well-established method for inducing intestinal ROS – one of the key physiological responses to enteric infection and inflammation. In the revised Discussion, we explicitly acknowledge that pathogenic infections – which trigger both intestinal ROS and additional immune pathways – may engage distinct or complementary mechanisms compared to chemically induced oxidative stress. We emphasize the importance of future studies aimed at dissecting these differences. In fact, we are actively pursuing this direction in ongoing work examining sleep responses to enteric infection. For the purposes of the present study, however, we chose to focus on a tractable and specific model of ROS-induced stress to define the contribution of Unpaired cytokine signaling to gut-brain communication and sleep regulation. This approach allowed us to isolate the effect of oxidative stress from other confounding immune stimuli and identify a glia-mediated signaling mechanism linking gut epithelial stress to changes in sleep behavior.

(9) To confirm that animals lacking EEC Unpaired signaling are not more susceptible to ROS-induced damage, the authors assessed the survival of upd2 and upd3 knockdowns

on 1% H₂O₂ and concluded they display no additional sensitivity to oxidative stress compared to controls. It may be useful to include other tests of sensitivity to oxidative stress, in addition to survival.

We appreciate the reviewer's suggestion. In our view, survival is a highly informative and stringent readout, as it reflects the overall physiological capacity of the animal to withstand oxidative stress. Importantly, our data show that animals lacking EEC-derived Unpaired signaling do not exhibit reduced survival following H₂O₂ exposure, indicating that their oxidative stress resistance is not compromised. Furthermore, we previously confirmed that feeding behavior is unaffected in these animals, suggesting that their ability to ingest food (and thus the stressor) is not impaired. As a molecular complement to these assays in response to this point and others, we have also performed an assessment of neuronal apoptosis (a TUNEL assay, Fig. S3f,g). This assay did not identify an increase in cell death in the brains of animals fed peroxide-containing medium. Thus, gross neurological health, behavior, and overall survival appear to be resilient to the environmental treatment regime we apply here, suggesting that the outcomes we observe arise from signaling per se.

(10) The authors confirmed that animals lacking EEC-derived upd3 displayed sleep suppression similar to controls in response to starvation. These results led the authors to conclude that there is a specific requirement for EEC-derived Unpaired signaling in responding to intestinal oxidative stress. However, they previously showed that EEC-specific knockdown of upd3 and upd2 led to increased daytime sleep under normal feeding conditions. Their interpretations of their data are inconsistent.

We appreciate the reviewer's comment. While animals lacking EEC-derived Unpaired signaling show increased baseline sleep under normal feeding conditions, they still exhibit a robust reduction in sleep when subjected to starvation – comparable to that of control animals (Fig. S3h–j). This demonstrates that they retain the capacity to appropriately modulate sleep in response to metabolic stress. Thus, the sleep-promoting phenotype under normal conditions does not reflect a generalized inability to adjust sleep behavior. Rather, it highlights a specific role for Unpaired signaling in mediating sleep responses to intestinal oxidative stress, not in broadly regulating all sleep-modulating stimuli.

(11) The authors report a significant increase in JAK-STAT activity in surface glial cells at ZT0 in animals fed 1% H₂O₂-containing food for 20 hours. This response was abolished in animals with EEC-specific knockdown of upd2 or upd3. The authors confirmed there were no unintended neuronal effects on upd2 or upd3 expression in the heads. They also observed an upregulation of dome transcript levels in the heads of animals with EEC-specific knockdown of upd3 fed 1% H₂O₂-containing food for 15 hours, which they interpret to be a compensatory mechanism in response to low levels of the ligand. This assay is inconsistent with previous experiments in which animals were fed hydrogen peroxide for 20 hours.

We thank the reviewer for identifying this discrepancy. The inconsistency arose from a labeling error in the manuscript. Both the JAK-STAT reporter assays in glial cells and the dome expression measurements were performed following 15 hours of H₂O₂ feeding, not 20 hours as previously stated. We have now corrected this in the revised manuscript.

(12) The authors show that animals with glia-specific dome knockdown did not have decreased survival on H₂O₂-containing food, and displayed normal rebound sleep in the morning following sleep deprivation. These results potentially undermine the significance of the paper. If the normal sleep response to oxidative stress is an important protective mechanism, why would oxidative stress not decrease survival in dome knockdown flies (that don't have the normal sleep response to oxidative stress)? This suggests that the

proposed mechanism is not important for survival. The authors conclude that Dome-mediated JAK-STAT signaling in the glial cells specifically regulates ROS-induced sleep responses, which their results support.

We agree that our survival data show that glial dome knockdown does not reduce survival under continuous oxidative stress. However, we believe this does not undermine the importance of the sleep response as an adaptive mechanism. In our survival assay, animals were continuously exposed to 1% H₂O₂ without the opportunity to recover. In contrast, under natural conditions, oxidative stress is likely to be intermittent, and the ability to mount a sleep response may be particularly important for promoting recovery and maintaining homeostasis during or after transient stress episodes. Thus, while the JAK-STAT-mediated sleep response may not directly enhance survival under constant oxidative challenge, it likely plays a critical role in adaptive recovery under natural conditions.

(13) Altogether, the authors conclude that enteric oxidative stress induces the release of Unpaired cytokines which activate the JAK-STAT pathway in subperineurial glia of the BBB, which leads to the glial downregulation of receptors for AstA, which is a wake-promoting factor also released by EECs. This mechanism is supported by their results, however, this research raises some intriguing questions, such as the role of upd2 versus upd3, the role of AstA-R1 versus AstA-R2, the importance of this mechanism in terms of survival, the sex-specific nature of this mechanism, and the role that nutritional availability plays in the dual functionality of Unpaired cytokine signaling in regards to sleep.

We thank the reviewer for highlighting these important questions. Our data suggest that Upd2 and Upd3, while often considered partially redundant, both contribute to sleep regulation, with stronger effects observed for Upd3. This is consistent with prior studies indicating overlapping but non-identical roles for these cytokines. Similarly, although AstA-R1 and AstA-R2 can both be activated by AstA, knockdown of AstA-R2 consistently produces more robust sleep phenotypes, suggesting a predominant role in mediating this effect. The possibility of sex-specific regulation is indeed compelling. While our study focused on females, many gut hormones show sex-dependent activity, and we recognize this as an important avenue for future research. Finally, we have included new data in the revised manuscript showing that gut-derived AstA is downregulated under oxidative stress, further supporting our model in which Unpaired signaling suppresses arousal pathways during intestinal stress

(14) Data Availability: It is indicated that: "Reasonable data requests will be fulfilled by the lead author". However, eLife's guidelines for data sharing require that all data associated with an article to be made freely and widely available.

We thank the reviewer for pointing this out. We have revised the Data Availability section of the manuscript to clarify that all data will be made freely available from the lead contact without restriction, in accordance with eLife's open data policy.

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